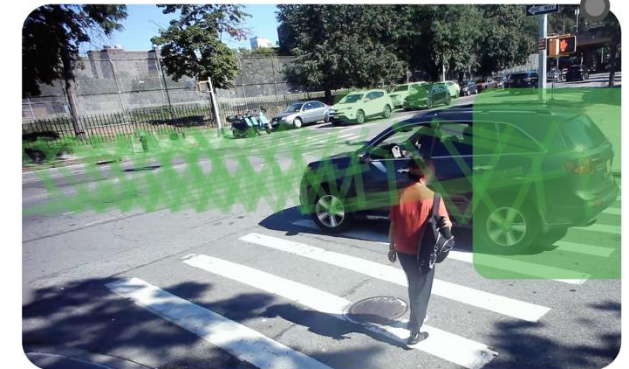
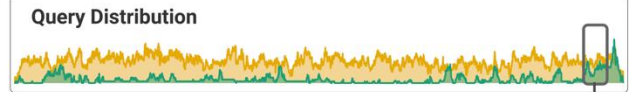
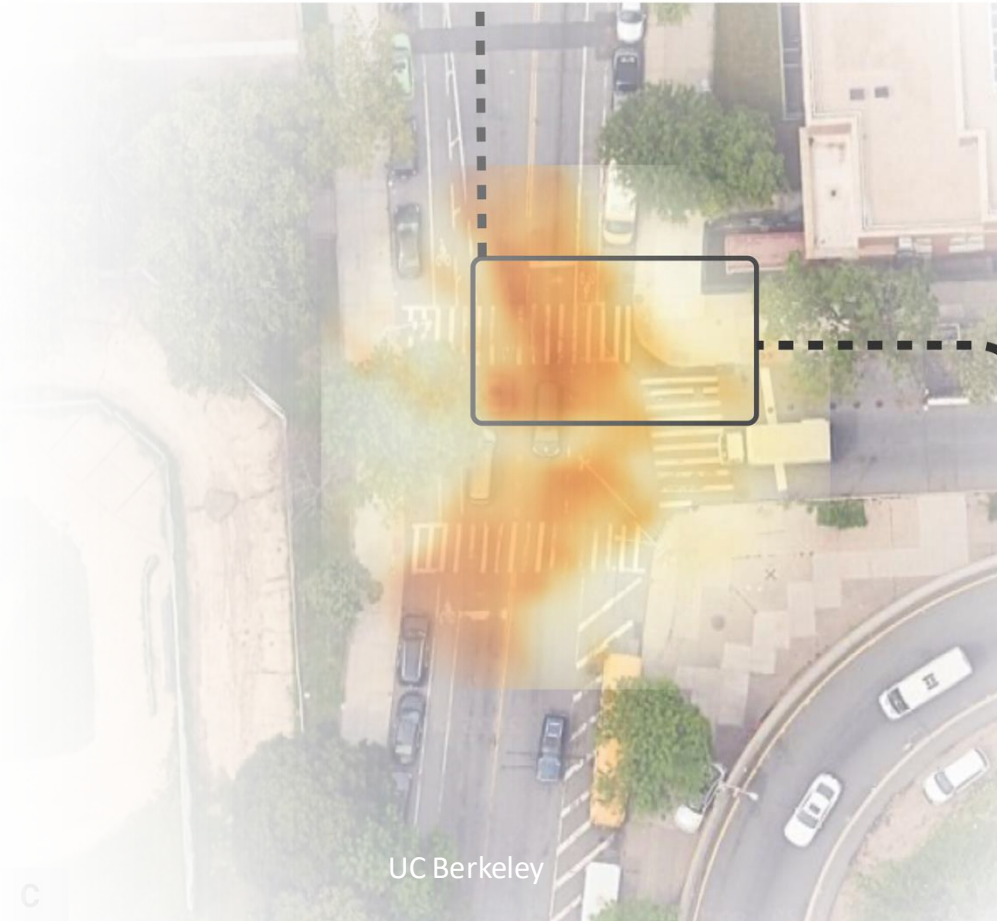
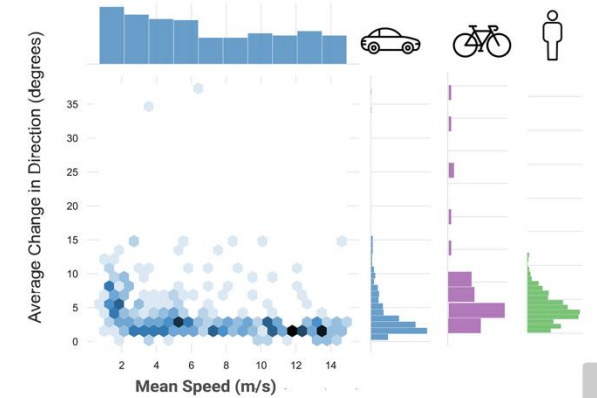
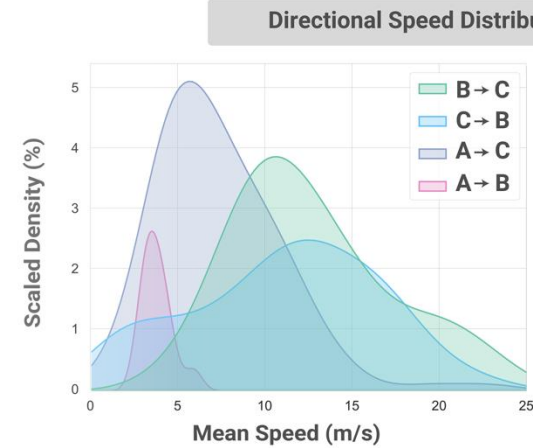
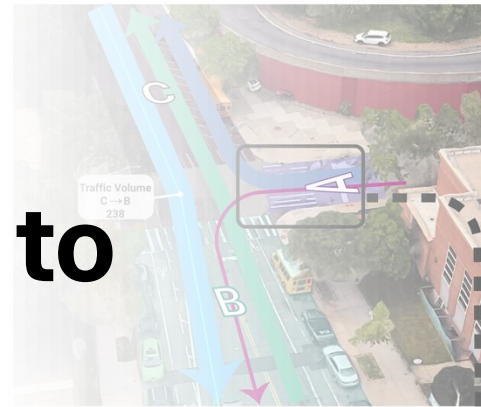


Introduction to Urban Data Analytics

Fall 2026

Maryam Hosseini

City & Regional Planning



UC Berkeley

C

B

D

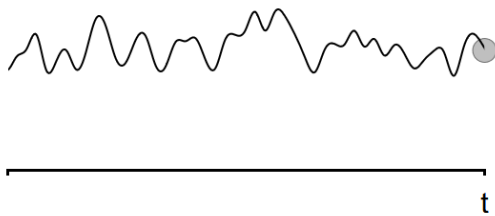
Module II:

Geospatial Analysis & Visualization

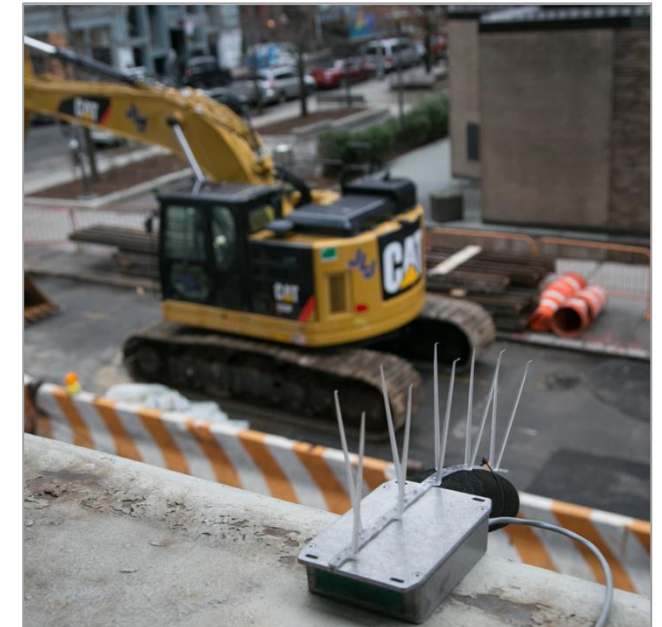
The background of the slide is a dark, textured world map. The map is composed of a dense network of small, light-colored icons and lines. These icons include various symbols such as people, houses, cars, and location pins, which are interconnected by a web of thin lines, suggesting a global network or data flow. The overall aesthetic is high-tech and data-driven.

The Expanding Landscape of Urban Data

Urban data

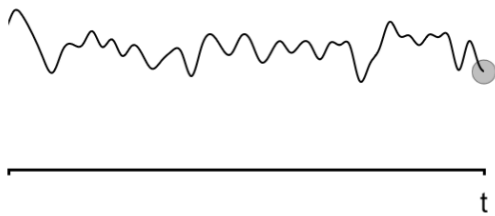


Array of Things (Chicago)



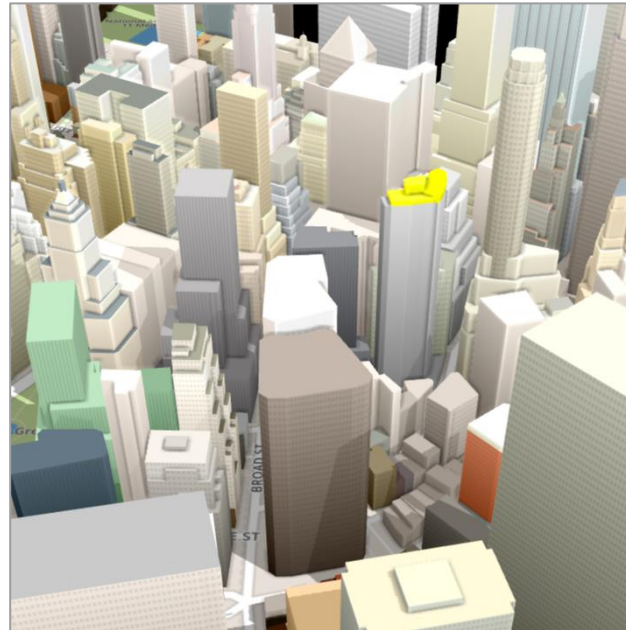
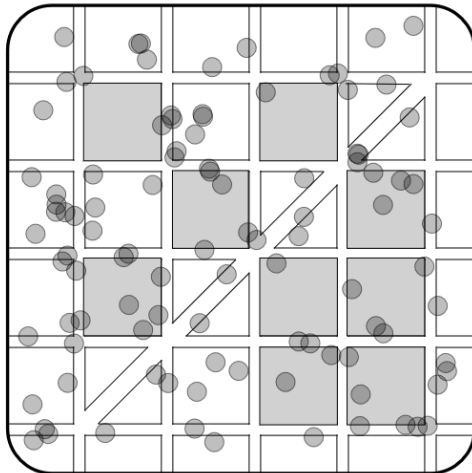
Sounds of New York City
(NYC)

Urban data



NYC
Taxi & Limousine
Commission

Urban data



Open Street Maps

NYC Information Technology & Telecommunications

한국어 > Translate | Text-Size

About Initiatives Residents Business For Agencies Contact Search

NYC 3-D Building Model

f t s t e Share Print

NYC 3-D Building Model

Building on the rich geospatial data already provided, NYC DoITT is releasing a three-dimensional (3-D) Building Massing Model of New York City. The 3-D model includes every NYC building present in the 2014 aerial survey, the capture from which the model was developed.

Using the Open Geospatial Consortium's **CityGML** specification as the basis, the NYC 3-D Building Massing Model was developed to a hybrid specification combining elements from Level of Detail (LOD) 1 and 2. Highlights of the model include the differentiation of building components including roof, facades and ground plane. All major roof structures are modeled. Pitched roofs are modeled, however domes and rounded roofs are not. All roof appendages are excluded including but not limited to chimneys, parapets, spindles and antenna. In addition, a subset of 'iconic' NYC buildings was modeled to LOD 2. Lastly the buildings are attributed with a Building Identification Number (BIN) to provide a means to associate other NYC data for analysis and rendering. For further details, see the included metadata.

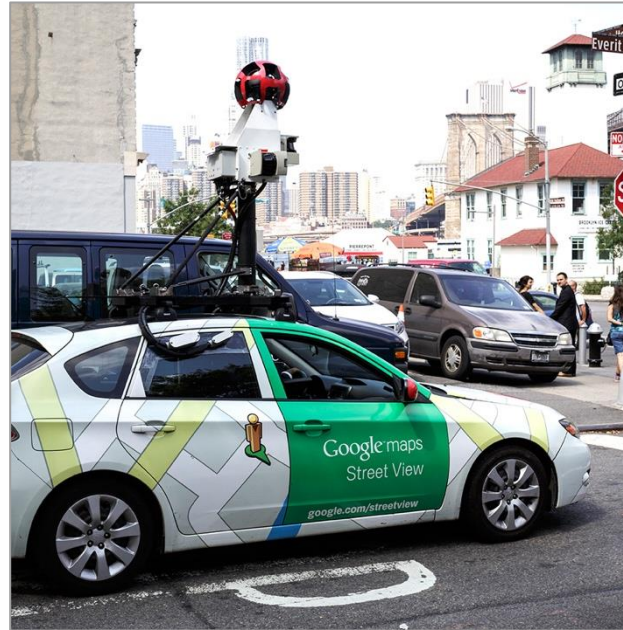
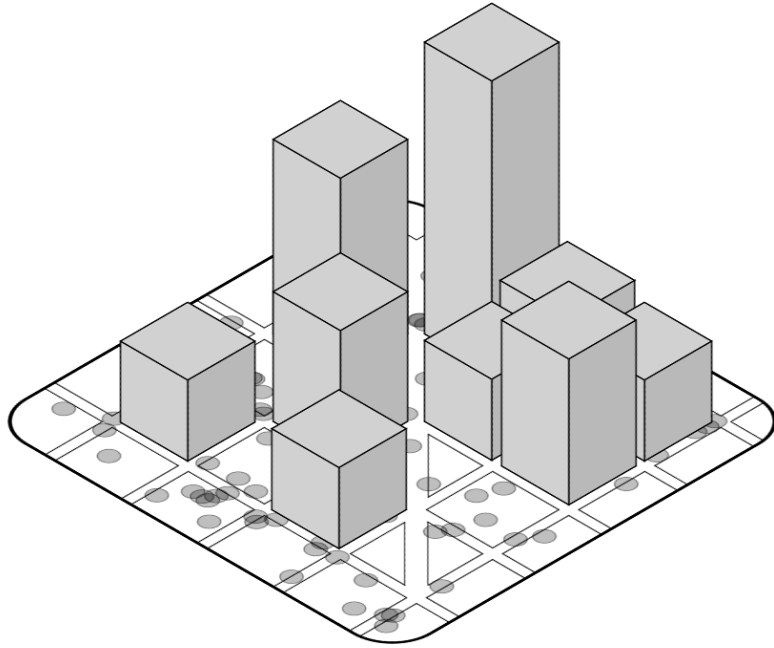
The 3-D model was a one-time capture. If the model proves useful, future updates and extension will be considered.

Download

Format	Size	URL
CityGML	894 MB	http://maps.nyc.gov/download/3dmodel/DA_WISE_GML.zip
Multipatch (ESRI)	251 MB	http://maps.nyc.gov/download/3dmodel/DA_WISE_Multipatch.zip

DoITT

Urban data



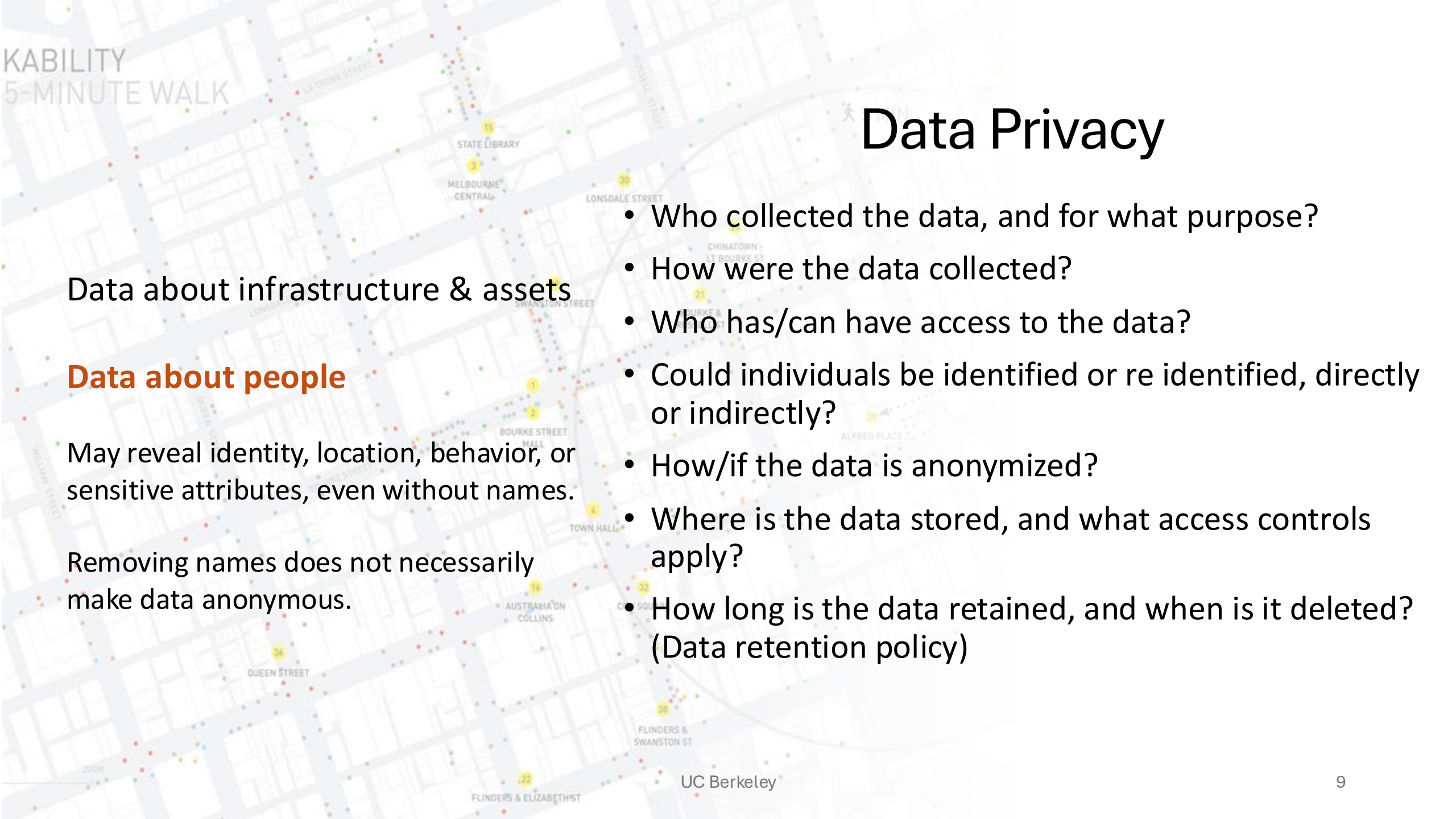
Google Street View



Urban Data

- Studying Urban Processes
- Analyzing Urban Change
- Identify & Address Urban Problems

- **Space + Time (Where & When?)**
- Surge in the availability of fine-scale data
- Varying spatio-temporal levels
- Understanding the data to choose the right one for our task, matters
- Every row in our datasets represents **people** or places, resources, and services that **matter in someone's life.**



Data Privacy

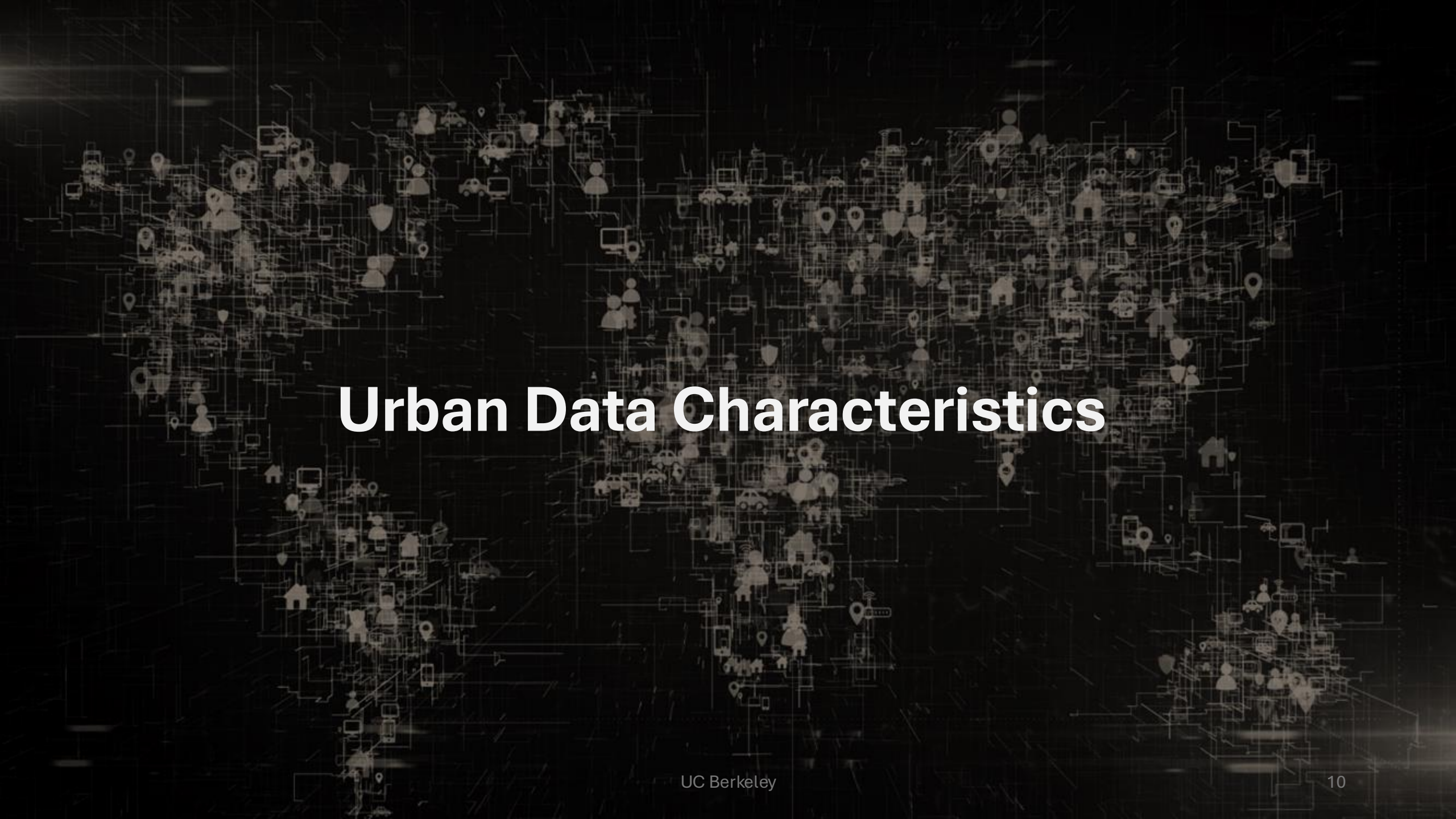
Data about infrastructure & assets

Data about people

May reveal identity, location, behavior, or sensitive attributes, even without names.

Removing names does not necessarily make data anonymous.

- Who collected the data, and for what purpose?
- How were the data collected?
- Who has/can have access to the data?
- Could individuals be identified or re identified, directly or indirectly?
- How/if the data is anonymized?
- Where is the data stored, and what access controls apply?
- How long is the data retained, and when is it deleted? (Data retention policy)



Urban Data Characteristics

Different Scales



Macro (City Scale)

Overall urban structure:
Population, Land Use,
Transportation Networks



Meso (Neighborhood Scale)

variation within the city:
Demographics, accessibility,
street networks, service
availability, development patterns



Micro (Human Scale)

Fine grained local conditions:
Sidewalks, curb ramps,
storefronts, pedestrian activity,
building characteristics

Spatial Data: Variation across Space

Spatial Data: Data that captures values across multiple locations and includes geographical identifiers.

- *Example: National unemployment rates broken down by state or city.*

Non-Spatial Data: Data without inherently spatial qualities.

If a dataset applies **uniformly** to a single fixed location, **it is still considered non-spatial** because there is **no variation** across space.

- *Example: A national unemployment rate for 2023 (single value for the whole country).*

Temporal Data: Variations across Time

Temporal Data: Data that represents values captured across multiple time-stamps.

- *Example: Annual national unemployment rate 2010-2025*

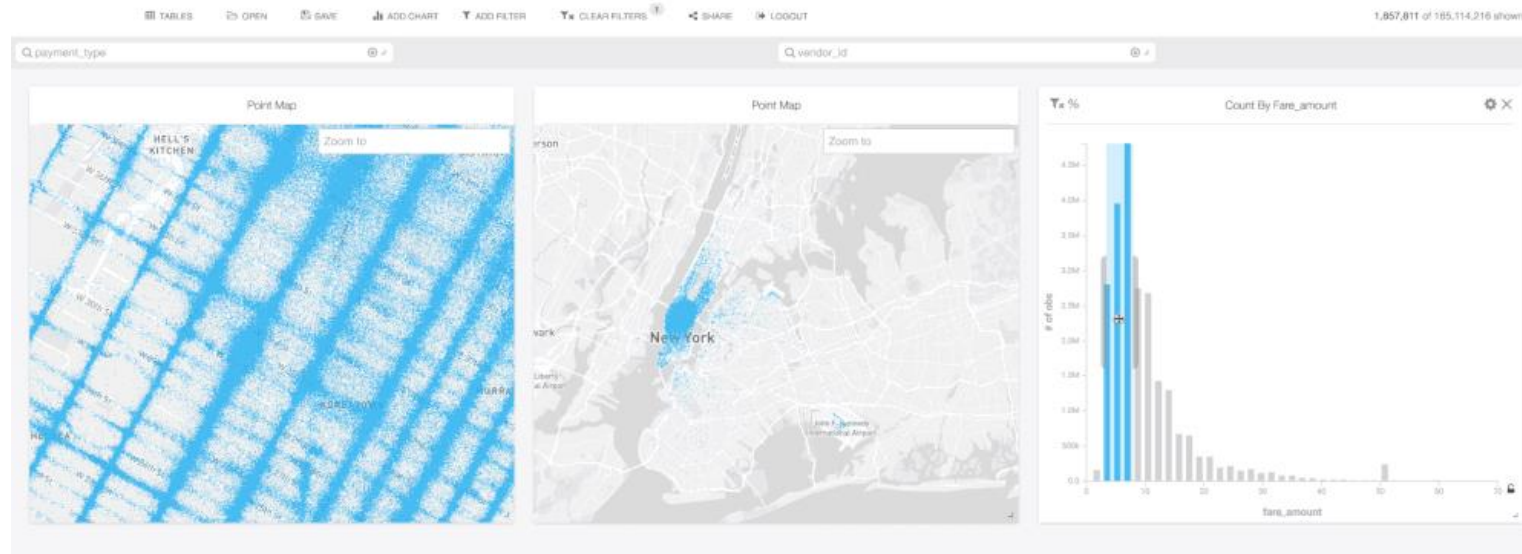
Non-temporal data (cross-sectional): Data that represents values captured at a specific point in time, hence, no variation across time.

- *Example: A national unemployment rate for 2023*

Spatio-temporal Data: Variation across time and space

Spatio-temporal Data: Data that represents values captured across various **locations** and **time**.

*Example: Annual unemployment rate of **EU countries** 2010-2025*



Granularity

- Level of detail or precision at which data is represented, collected, or analyzed,
- Determines the smallest unit of information included in the analysis.

- **Spatial**

(e.g., incident level, street level, city-level)

- **Temporal**

(e.g., hourly, daily, annual snapshots)



NYC PLUTO Landuse data - 2021. Source: [Morphocode](#)

MORPHOCODE

NYC Primary Land Use Tax Lot Output (PLUTO):

Spatial: **Parcel level**

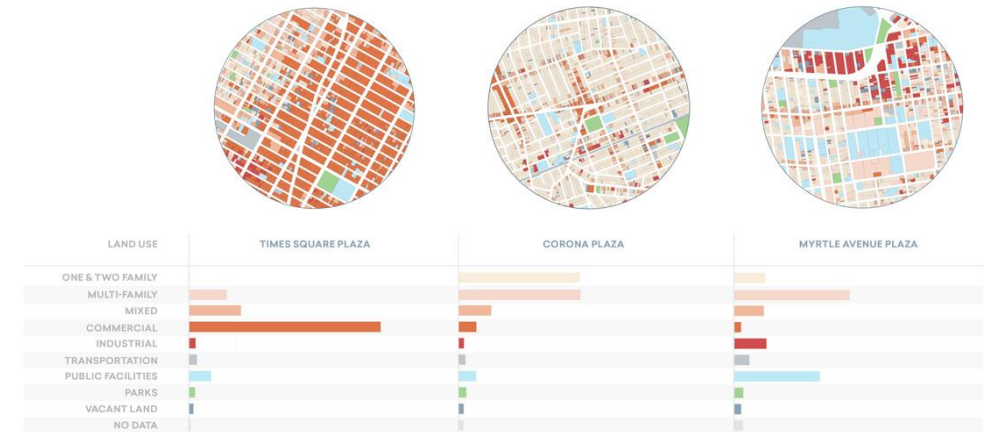
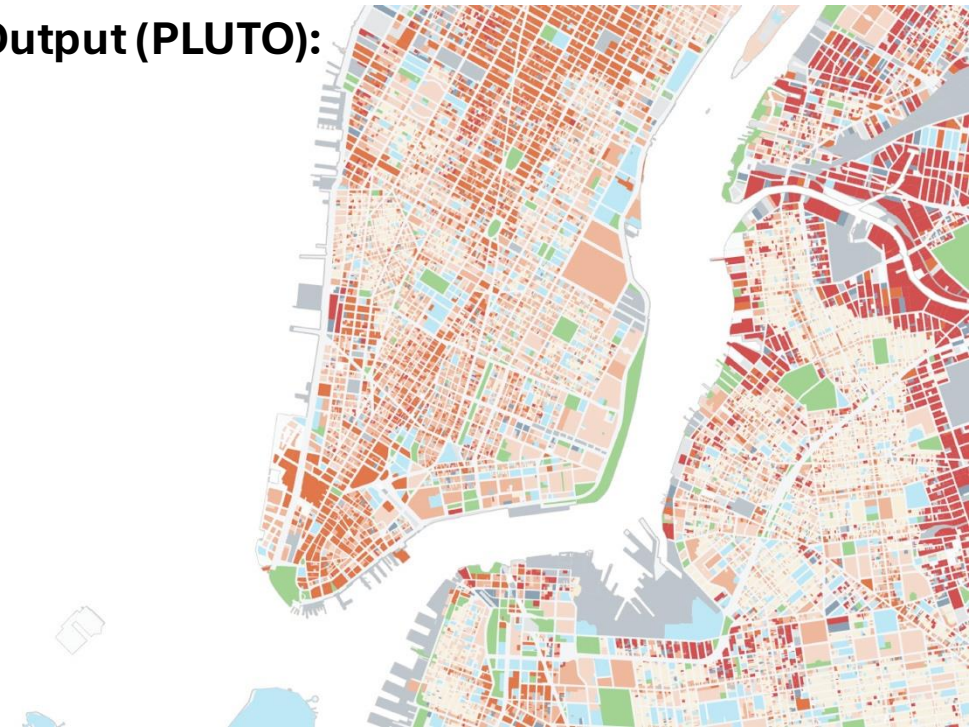
Temporal: **Quarterly**

Granularity

- **Decomposition:** How much detail is presented?
- **Abstraction:** What level of variation is being masked?

Example:

An annual dataset masks seasonal variations:
If a building's ownership changes multiple times within a year, only the last recorded owner is listed.



NYC PLUTO Landuse data - 2021. Source: [Morphocode](#)

MORPHOCODE

Granularity

- **Decomposition:** How much detail is presented?
- **Abstraction:** What level of variation is being masked?

Air pollution monitors often miss communities of color, study finds

By EHN Curators ▪ Mar 26, 2025 ▪ 2 min read



Print | PDF | Email



TOP NEWS

Without more granular, localized monitoring, the problem goes unnoticed, and residents in the affected area continue to face health risks.



Granularity

- **Decomposition:** Spatial and/or temporal resolution. How much detail is presented?
- **Abstraction:** What level of variation is being masked?

Example:

A city publishes an annual “ADA Sidewalk Compliance Score” by council district, reporting that 92% of sidewalks meet standards.

Masked changes:

District-level aggregation hides clusters of noncompliant curb ramps and surface problems around rehabilitation clinics and senior housings. Across the 8 adjacent blocks around a senior housing complex, <40% of crossings are usable.

Aggregation

- Going from more details to less details
- Aggregation reduces detail by grouping data and applying summary functions (mean, median, sum, count, rates)
- We aggregate on daily basis: our age, the time, location,
- Pedestrian signal times are based on an average able-bodied walking speed.
- Level of aggregation determines what changes we decide to mask (**abstraction**)

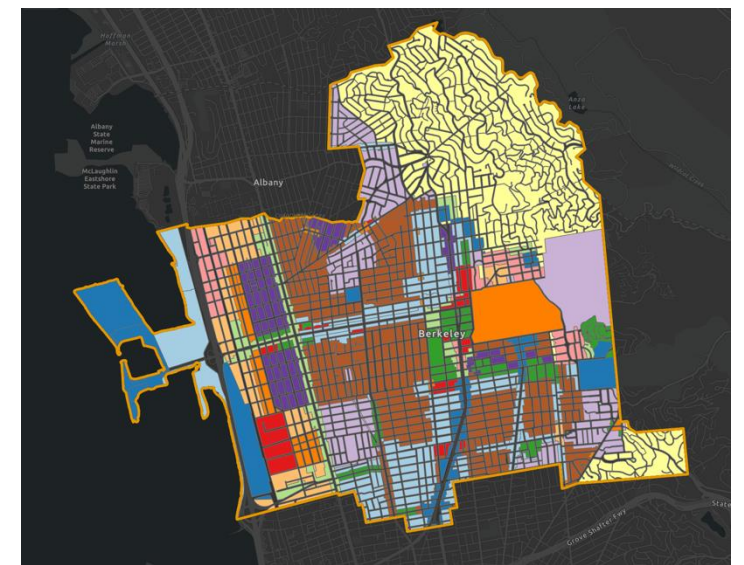




City



Neighborhoods



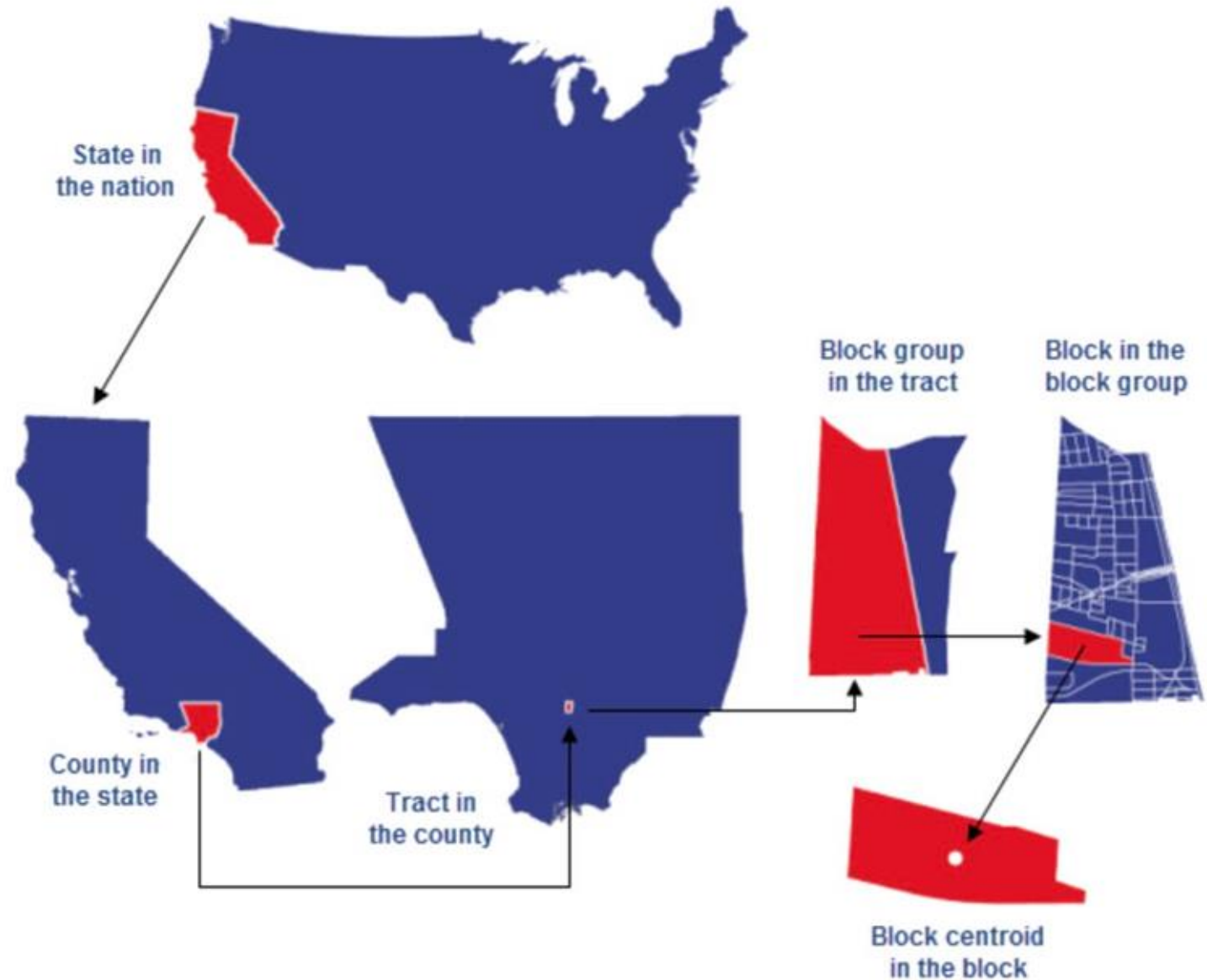
Zoning Districts

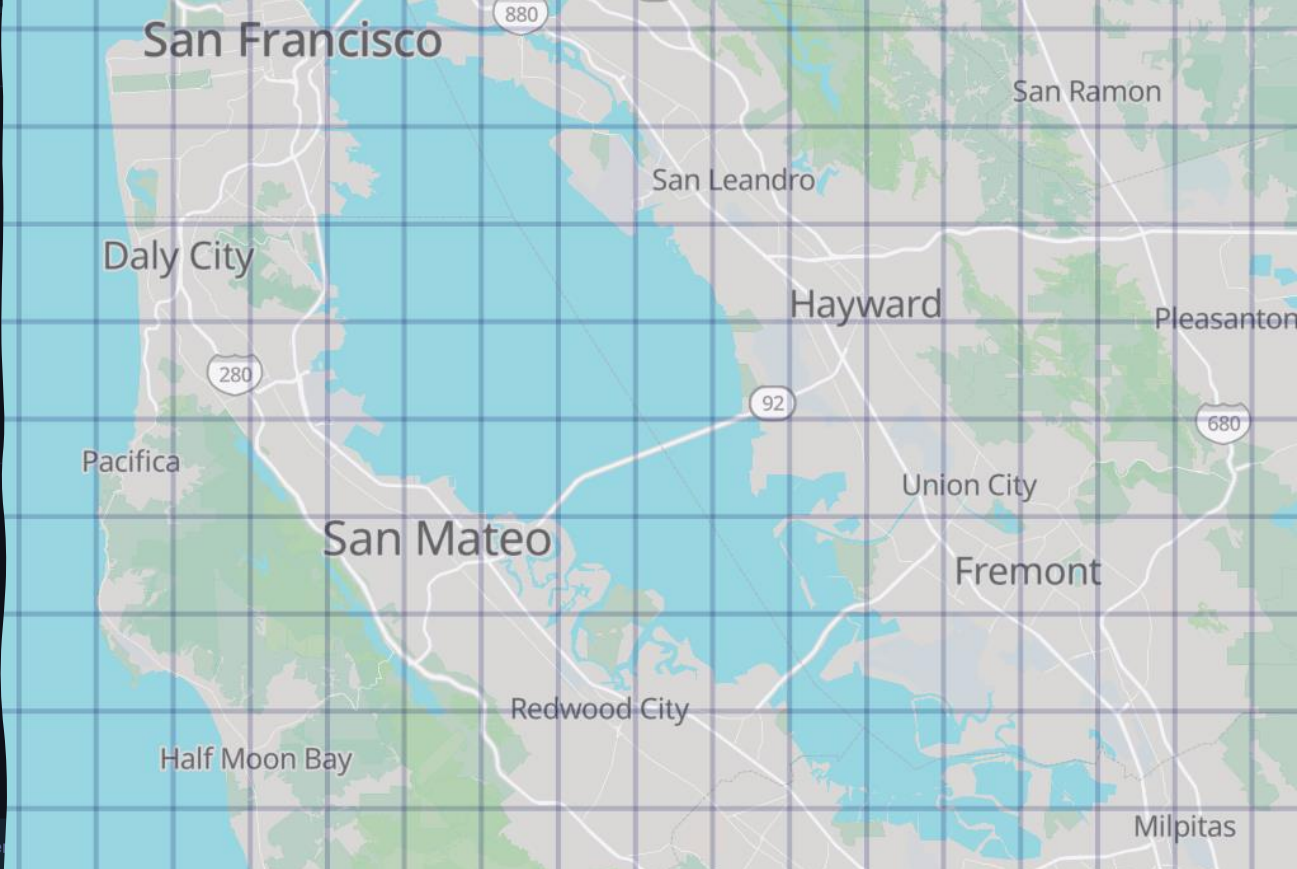
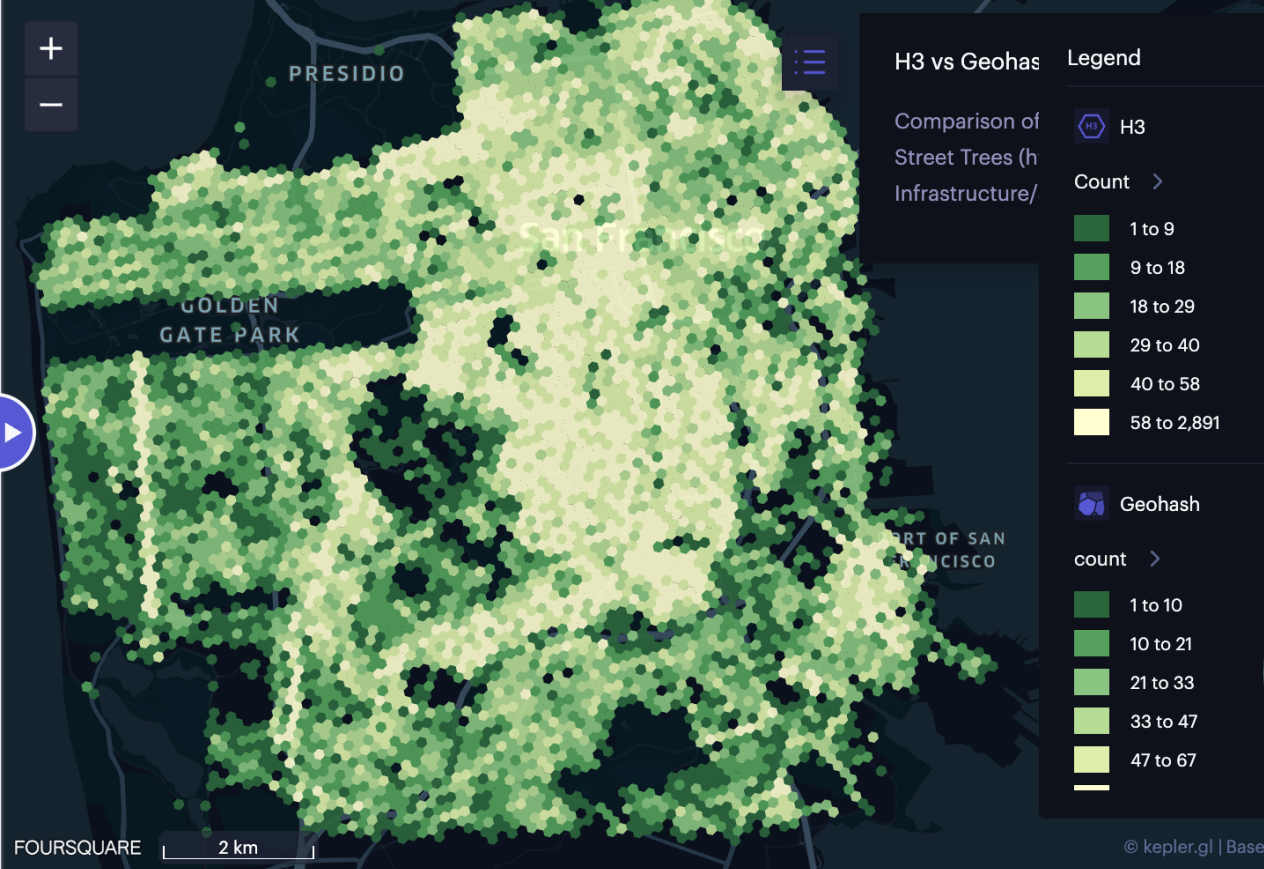
Administrative Boundaries

Government defined areas such as countries, states, counties, cities, districts, and neighborhoods.

Census Geographies

Geographic areas defined or used by the Census Bureau to collect, tabulate, and publish census data. Such as State, County, Tract, Block group, Block, Lot



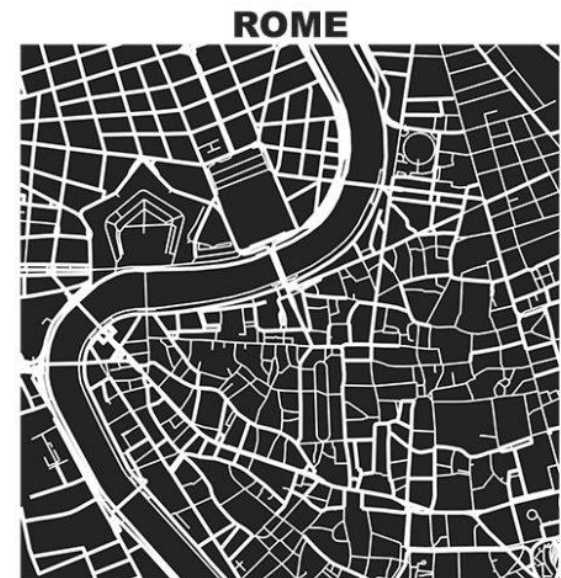


Grids

- Uniform cells, such as squares or hexagons, used to divide space independently of administrative boundaries.

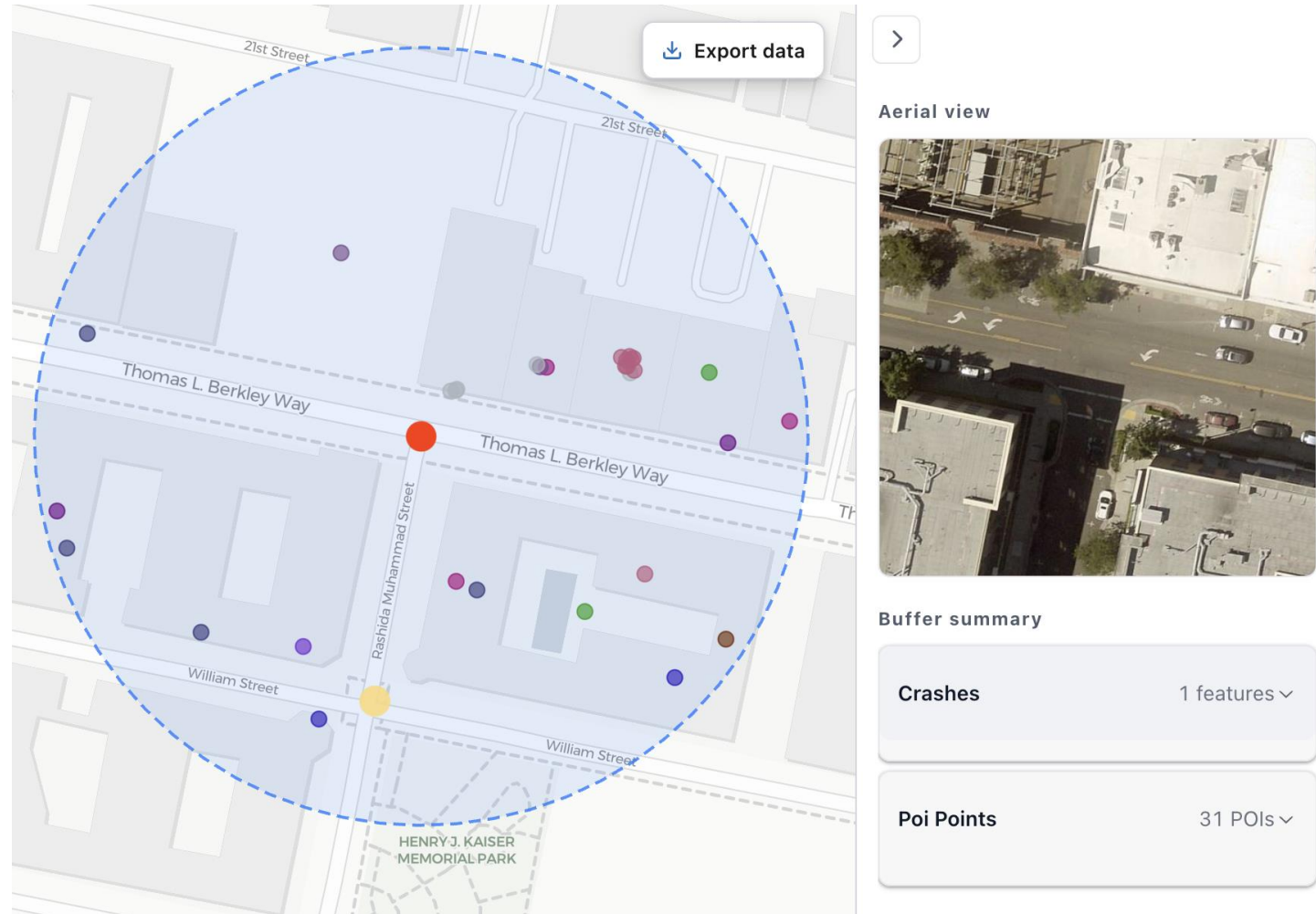
Streets

Street / segments used as spatial units for measures such as traffic, crashes, parking, or pedestrian activity.



Address Point

Individual locations that can be geocoded into coordinates for mapping and spatial analysis.



Data Type: Numerical Data

- **Discrete:** integers
 - Discrete data values have distinct, separate values.
 - The data type `int` in Python
 - Example: number of people, number of bikes
- **Continuous:** floating points
 - Continuous data values can fall anywhere within a range of measurements.
 - The data type `float` in Python
 - Example: height, precipitation, distance

Data Type: Categorical Data

Nominal:

- Nominal data are categories or named data with no inherent hierarchy or structure, they explain the category without any order
 - Example: **Transportation Modes** (Bus, Train, Bicycle, Car, Walking)

Ordinal

- Ordinal data demonstrates a specific order, hierarchy or known structure through the categorical names
 - Example: **Housing Affordability Categories** (Affordable, Moderate, Expensive, Very Expensive)

Urban Data Formats



Tabular: Structured data in a **table format** (rows & columns).

Examples: CSV, Excel, relational databases.



Raster (image): **Grid-based** data where each cell (pixel) has a value.

Examples: Images (JPEG, PNG, TIFF), Satellite Data, Digital Elevation Models (DEMs).

Video: Temporal raster data: a sequence of raster images over time.

Examples: MP4, AVI, drone videos



Vector: **Geospatial representations** using **points, lines, polygons**.

Examples: Shapefiles (SHP), GeoJSON, OpenStreetMap data.

Urban Data Formats



Graph: Nodes and Edges representing relationships/networks.

Examples: Social networks, transportation networks, citation networks.



3D Geometry - Explicit 3D shapes and surfaces used in modeling.

Examples: CAD models, 3D meshes (OBJ, STL), BIM models.



LiDAR Point Cloud - Sparse 3D spatial data collected via LiDAR sensors.

Examples: LAS, LAZ (georeferenced elevation/terrain scans).



Sound Data - Audio signals stored as waveforms or spectral data.

Examples: WAV, MP3, FLAC (raw or processed sound recordings).

Assessing data quality

- Looking at all the details provided about the data:
 - **Metadata**, or “data about the data”
 - Time and date of creation
 - The context in which the data is presented
- For online datasets, you can often find this information by navigating to the “About” or “More Information” web pages or by following a “Documentation” or “Data Dictionary” link.

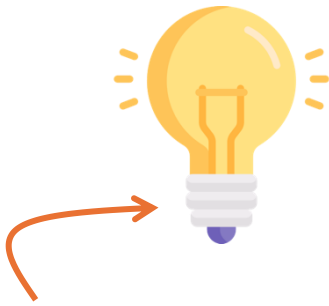
Who Does Your Data Represent?

- **Be precise about your population and time.** Do not generalize from a specific sample of users to the broader public.
- Example: If you analyze Reddit data about e-bikes, you are **not analyzing “public perception of micromobility.”** You are analyzing *how **Reddit users** discussed or perceived **e-bikes** during a specific **time period**.*
- Document **why** the data were collected, **what** they contain, **how** they were collected, their **limitations**, and appropriate/inappropriate uses.

Data Sources

Category	Examples	Data Provided	Use Cases
Official Government (Federal, State, Local)	Census Bureau, DOT	Population, transportation, climate	Urban planning, policy analysis, infrastructure development
	EPA, NOAA	Pollution, environmental risks	Climate resilience, sustainability
Crowdsourced & Open	OpenStreetMap (OSM)	Roads, sidewalks, POIs	Accessibility mapping, navigation
	Project Sidewalk	Sidewalk accessibility	ADA compliance, walkability studies
Proprietary & Commercial	Google Maps API, HERE	Traffic, geolocation, POIs	Mobility analysis, retail planning
	Placer.ai, StreetLight Data	Human movement, GPS-based mobility	Transportation modeling, economic insights

Describing your Data



Final Project
Midterm / Final Submissions

- **Coverage** (*What is the spatial and temporal extent?*)
 - Geographic Coverage (e.g., "New York City, parcel-level")
 - Time Range (e.g., "2010–2023, monthly")
- **Granularity** (Level of detail in the data)
 - Spatial: Parcel-level, block-level, citywide, etc.
 - Temporal: Hourly, daily, annual snapshots, etc.
 - Entity: Individual records vs. aggregated data.
- **Type** (*Nature of the data*)
 - Numerical (discrete or continuous)
 - Categorical (nominal or ordinal)
- **Format** (*How the data is stored or structured?*)
 - Tabular (CSV, Excel, SQL),
 - Geospatial (Shapefile, GeoJSON, Raster),
 - Time-series, Graph, Text, Point Cloud, etc.
- **Source** (*Where does the data come from?*)
 - Official agency, sensor-based, research dataset, proprietary, or crowdsourced.

Now what can be done with all
this data?

~~Now what can be done with all
this data?~~

What problems can I help solve?

What data do I need to tackle that?

Is that data available?

Can I help make that available?

Disaster management

Environmental Science

Accessibility

Public Health

Urban Planning

Human Rights

Safety How can I apply advanced
techniques to address real
challenges in ...

Social Well-fare

Environmental Justice

Access to Biodiversity

Transportation

Poverty

Data-informed not Data-driven!

- **Start from the question**, not the data
- Data should inform your decisions, not drive them.
- Our role is to identify a problem first and then determine what data, analysis methods, or visualization techniques are needed to address it.
- If the data falls outside your domain expertise, **work with or consult domain experts** before interpreting or analyzing it.

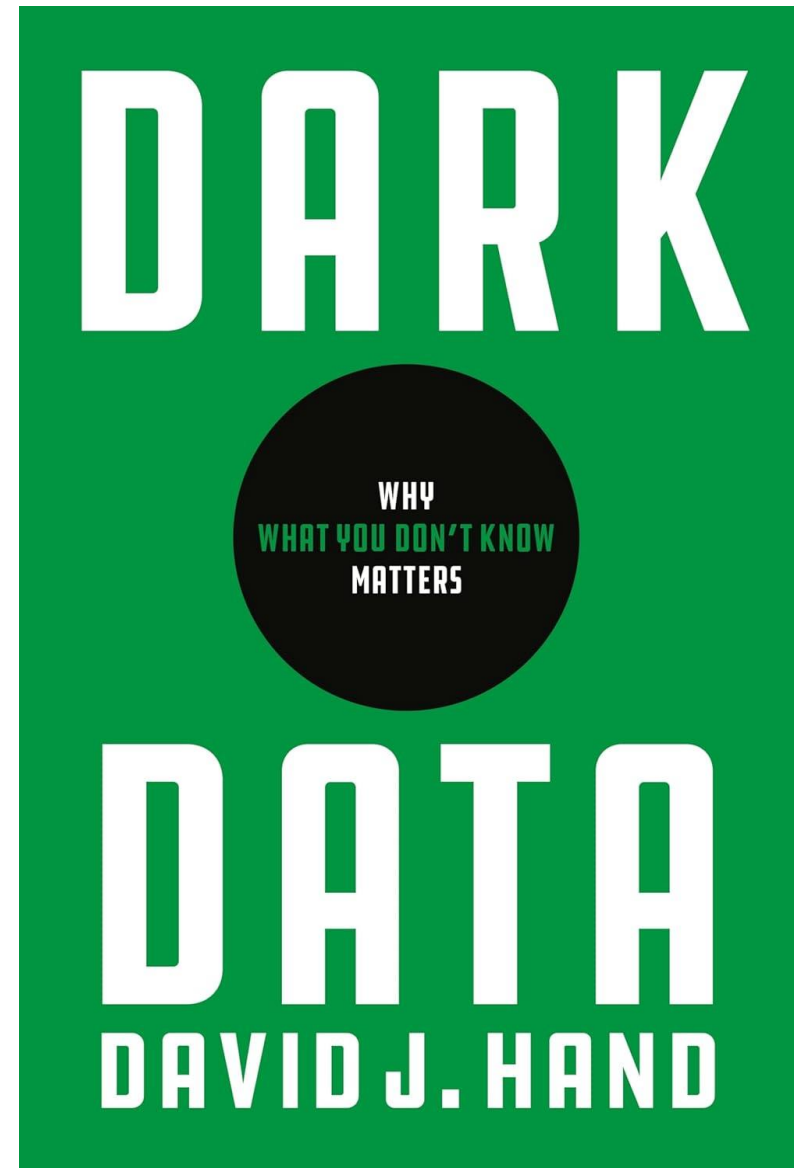
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Dark Data:

Why What You Don't Know Matters

- *DD- Type 1: Data We Know are Missing.*
- *DD- Type 2: Data We Don't Know are Missing*
- *DD- Type 3: Choosing Just Some Cases*
- *DD- Type 4: Self- Selection*
- *DD- Type 5: Missing What Matters*

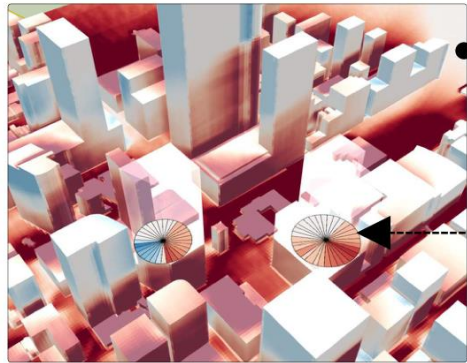


Quantification Trap

- The mistake of treating a numerical measure or proxy as if it were the phenomenon itself, and **assuming that improving the measure necessarily means improving what we actually care about.**
- Numbers help us simplify complex realities so we can compare, communicate, and make decisions.
- The measure is still a simplification. It leaves out context, judgment, and parts of the phenomenon that are difficult to quantify.
- The problem starts when we forget that distinction and begin treating the metric as the thing itself.
- **Then improving the number can become the goal**, even when the underlying situation has not actually improved.

Urban Data in Action

a



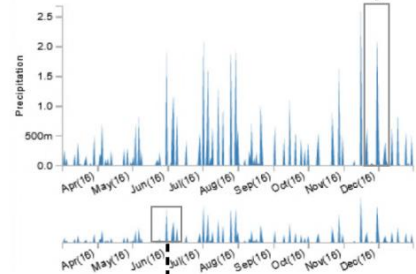
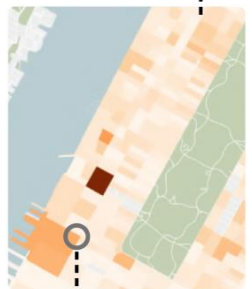
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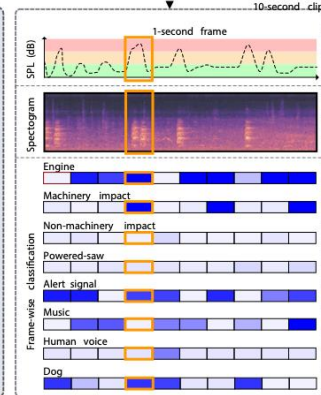
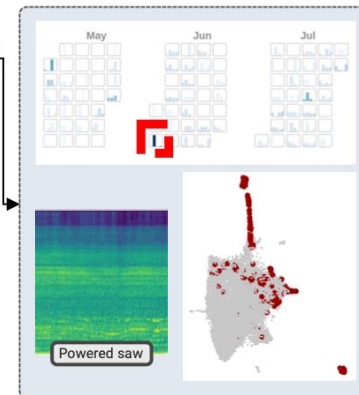
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b

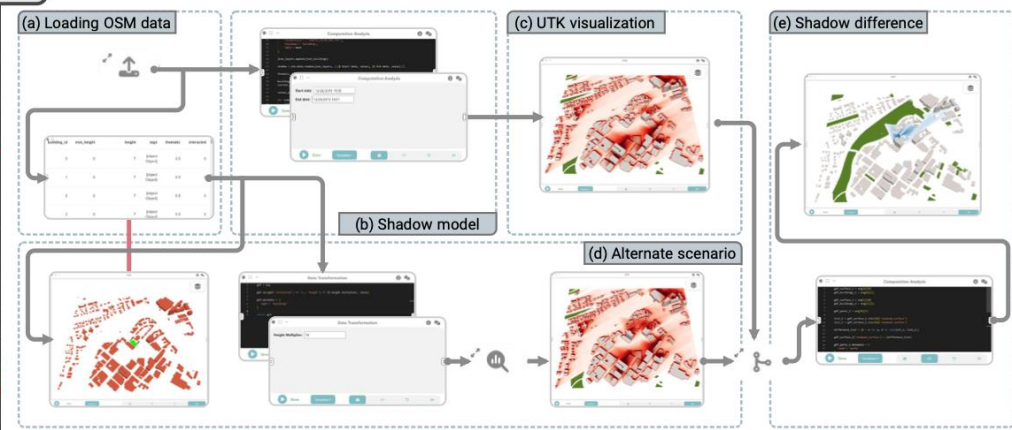


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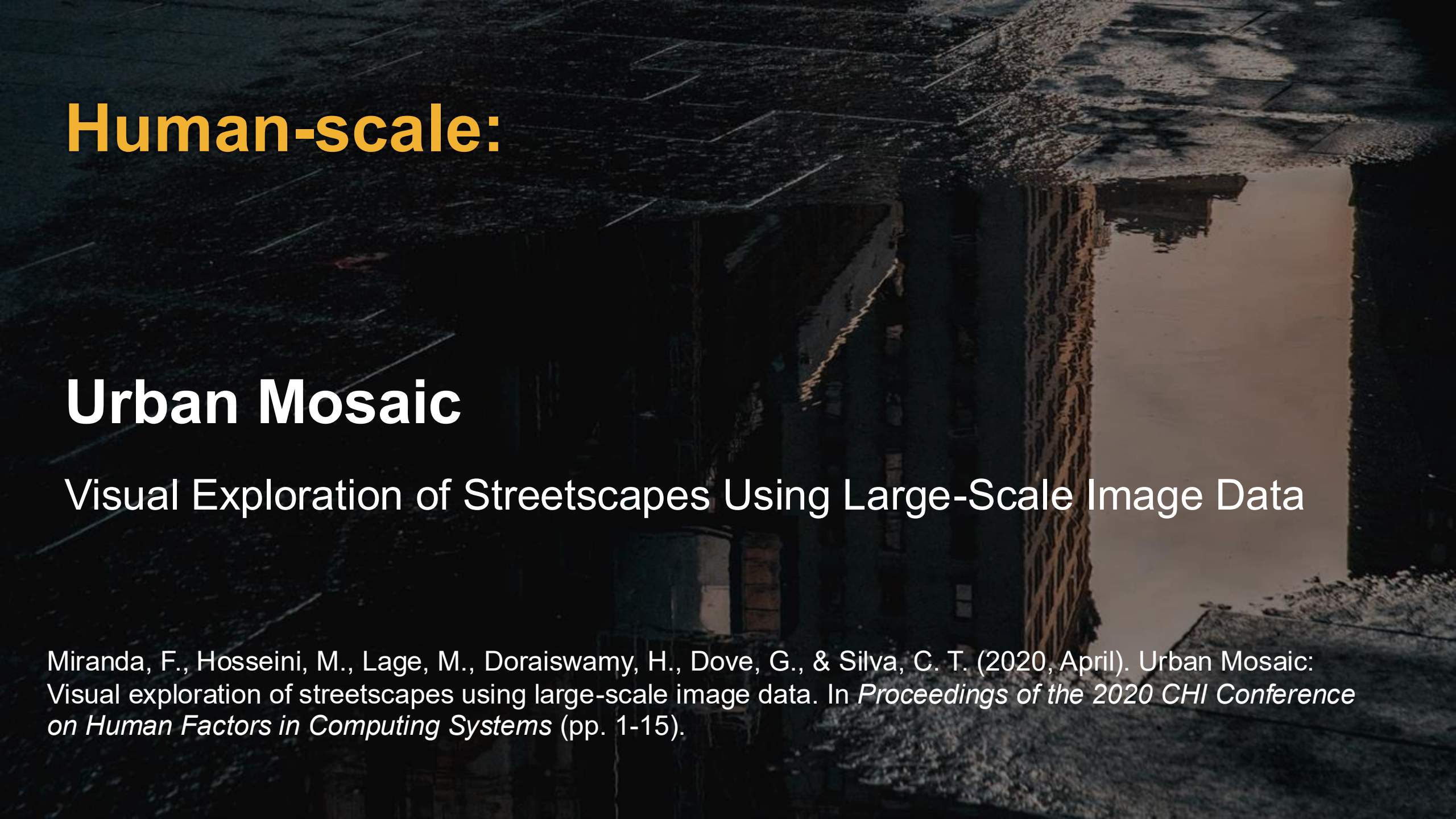


Querying by audio sample

d



a) Urban Toolkit b) Urban Mosaic c) Urban Rhapsody d) Curio

A dark, wet street at night. A large, clear reflection of a building is visible in a puddle on the right side of the image. The text is overlaid on the left side of the image.

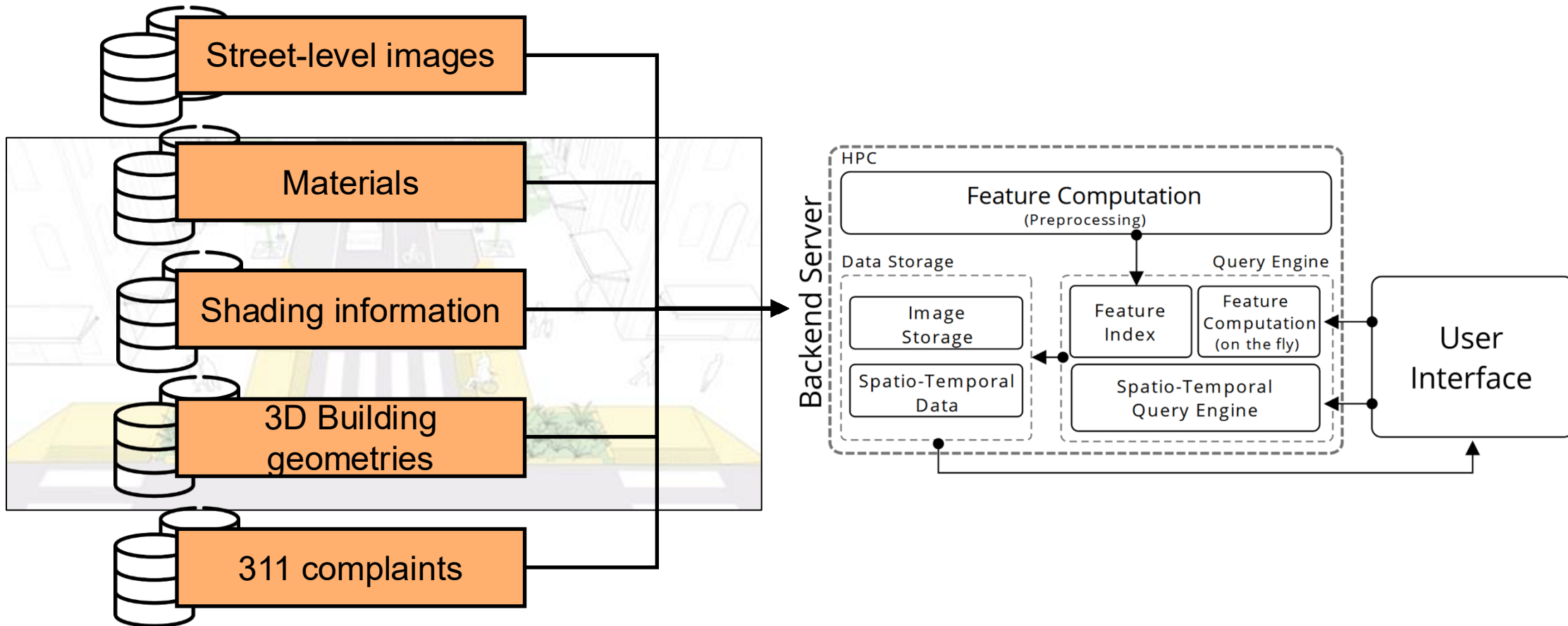
Human-scale:

Urban Mosaic

Visual Exploration of Streetscapes Using Large-Scale Image Data

Miranda, F., Hosseini, M., Lage, M., Doraiswamy, H., Dove, G., & Silva, C. T. (2020, April). Urban Mosaic: Visual exploration of streetscapes using large-scale image data. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1-15).

Urban Mosaic

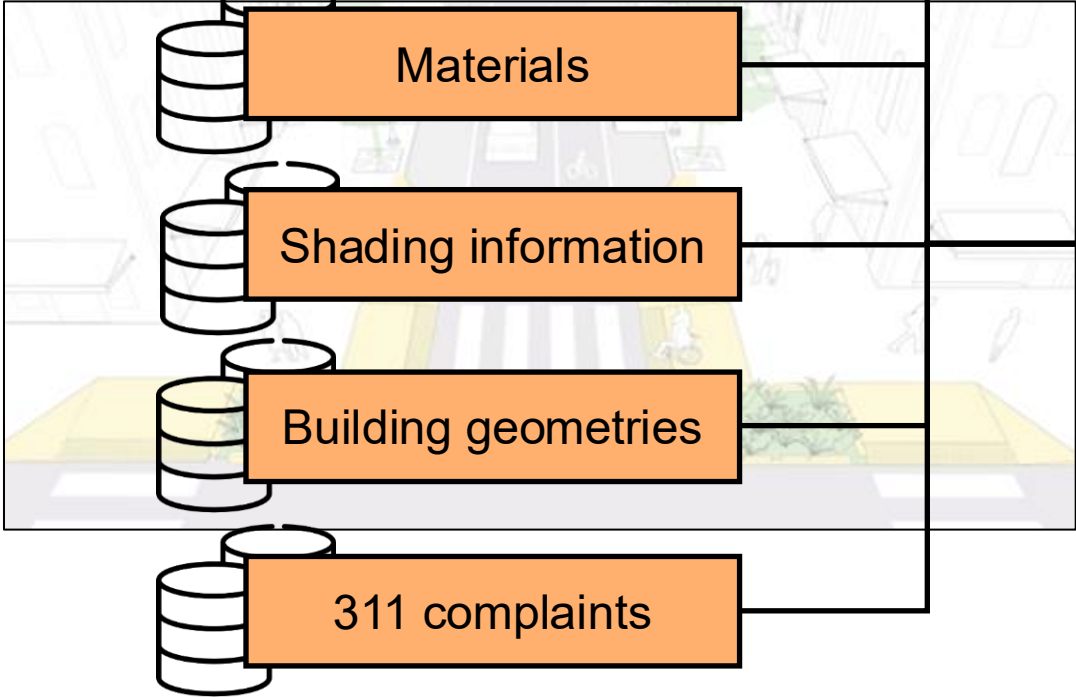


Materials

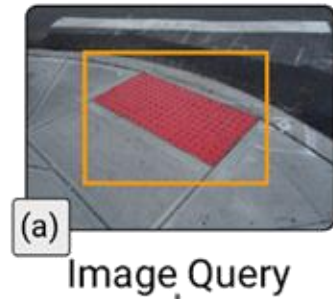
Shading information

Building geometries

311 complaints



Urban Mosaic



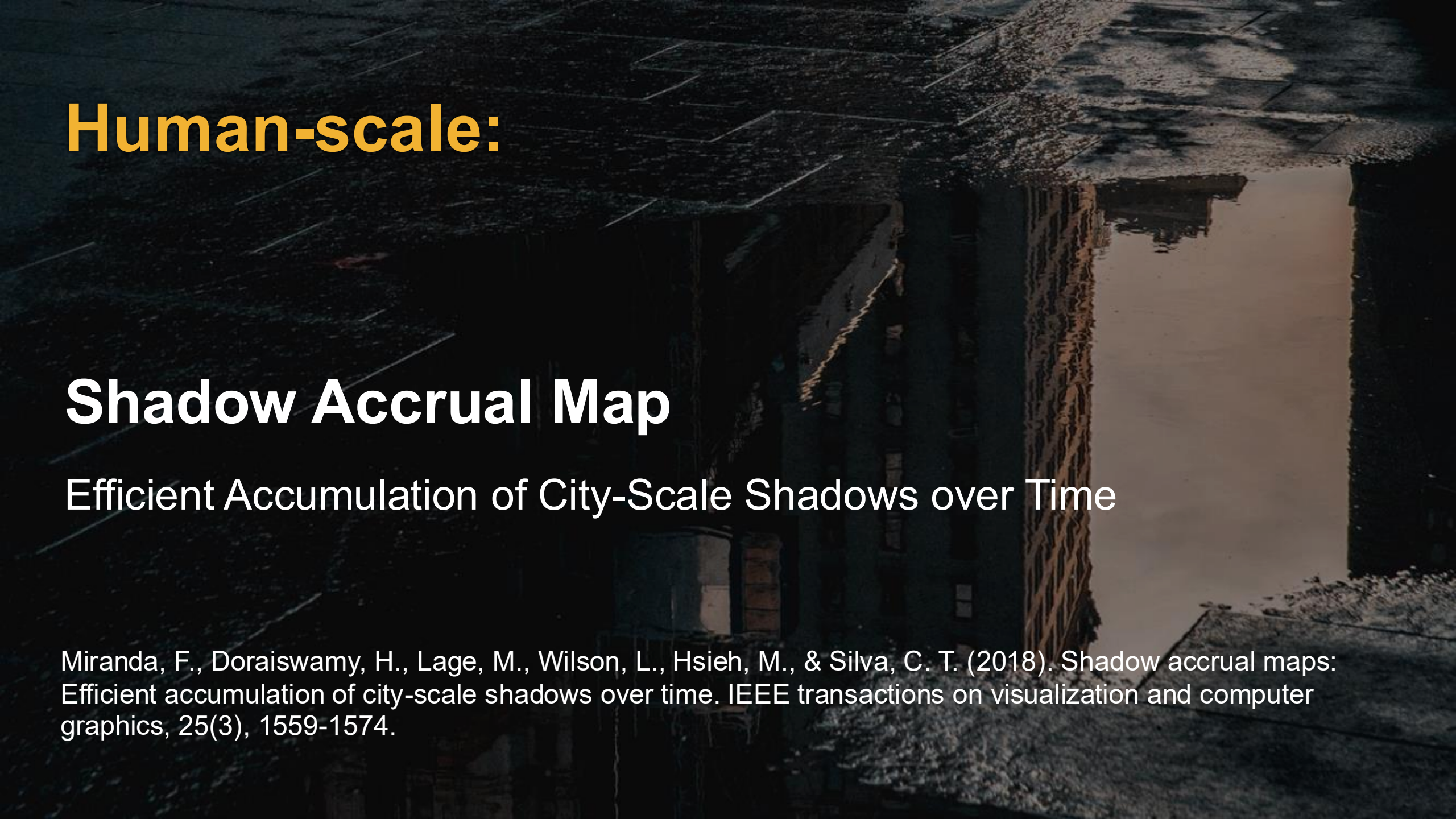
Site Evolution

Tracing the presence and condition of tactile paving in NYC

Urban Mosaic



Tracing the presence and condition of tactile paving in NYC



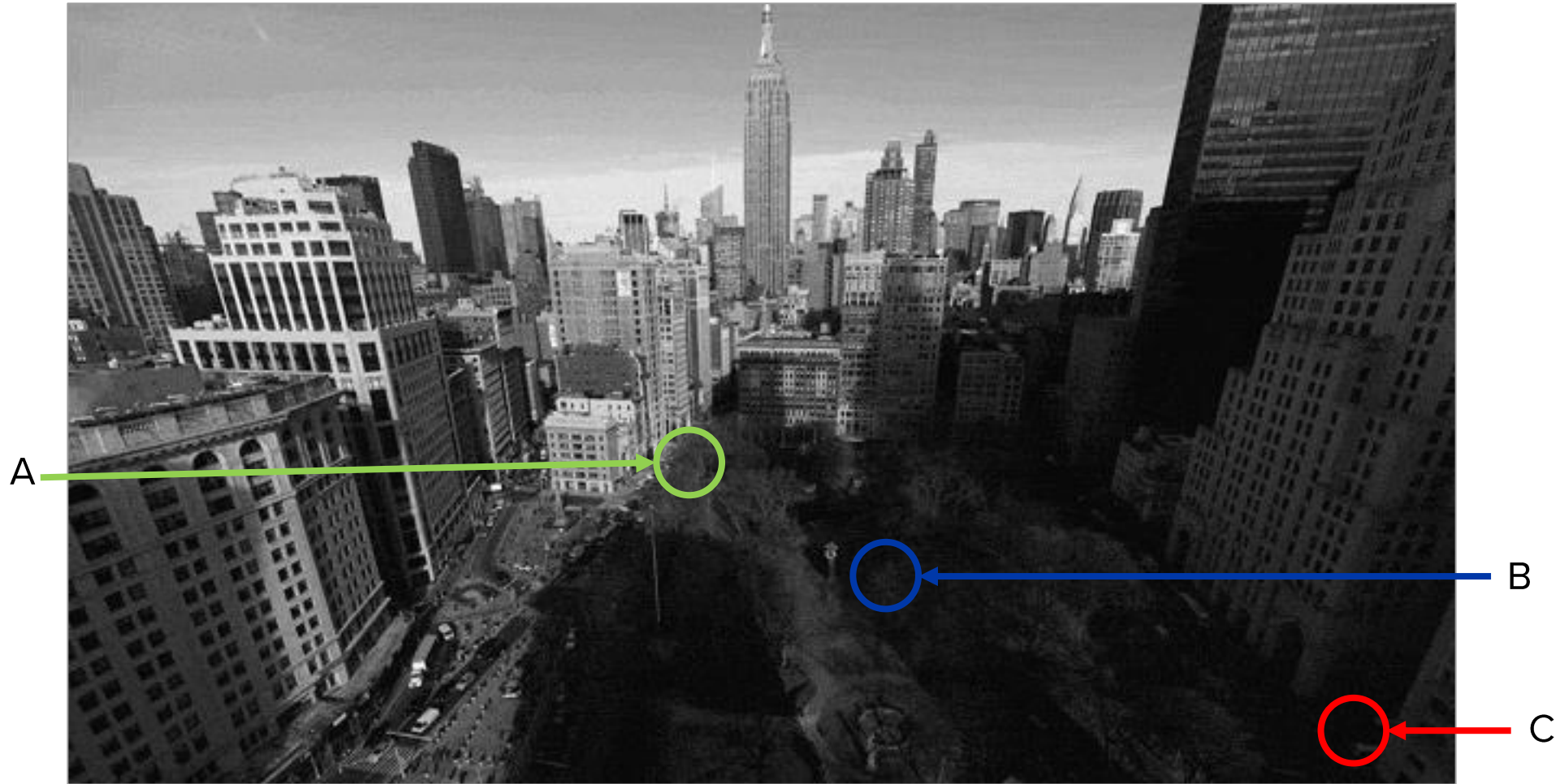
Human-scale:

Shadow Accrual Map

Efficient Accumulation of City-Scale Shadows over Time

Miranda, F., Doraiswamy, H., Lage, M., Wilson, L., Hsieh, M., & Silva, C. T. (2018). Shadow accrual maps: Efficient accumulation of city-scale shadows over time. *IEEE transactions on visualization and computer graphics*, 25(3), 1559-1574.

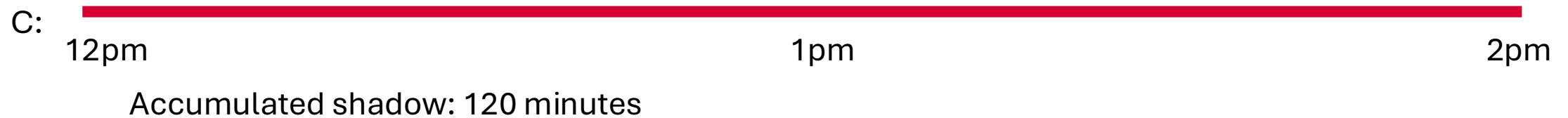
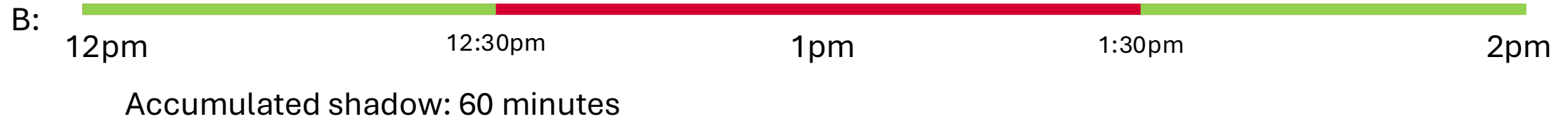
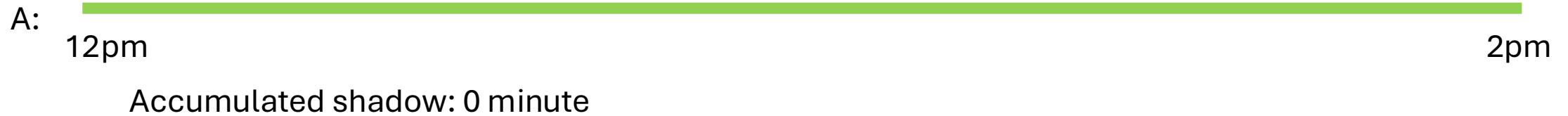
Quantifying shadow



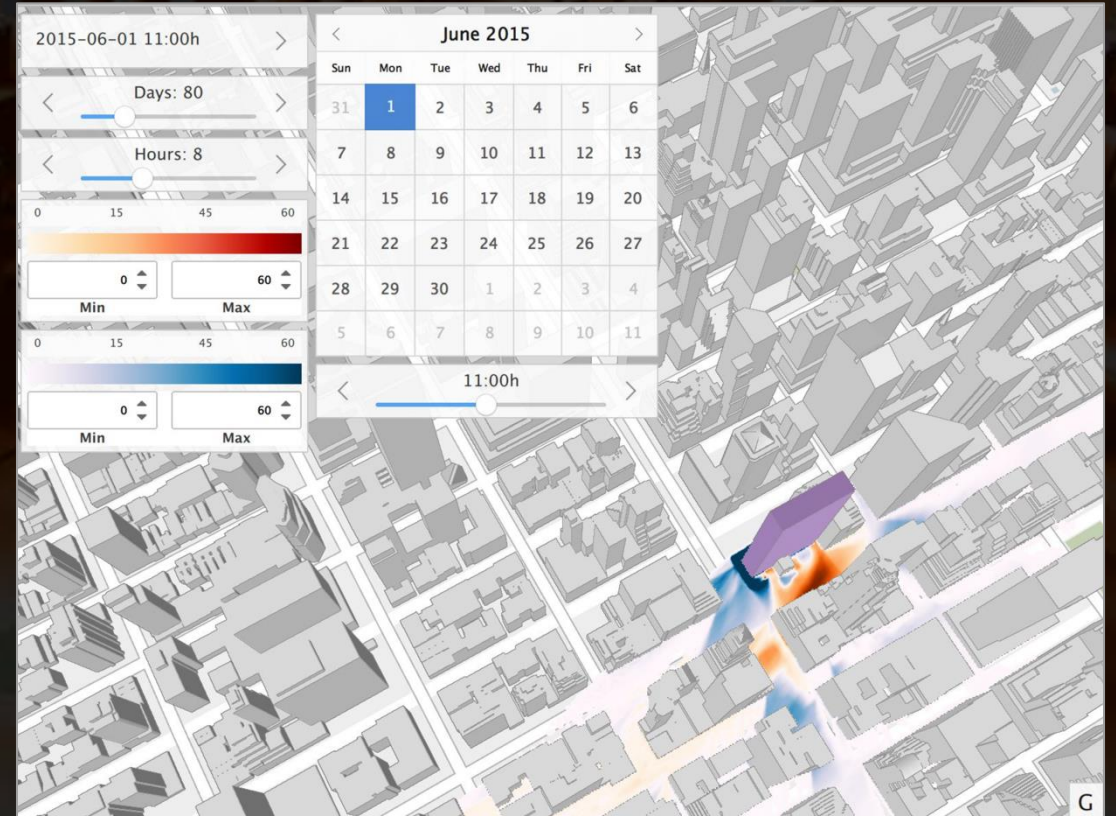
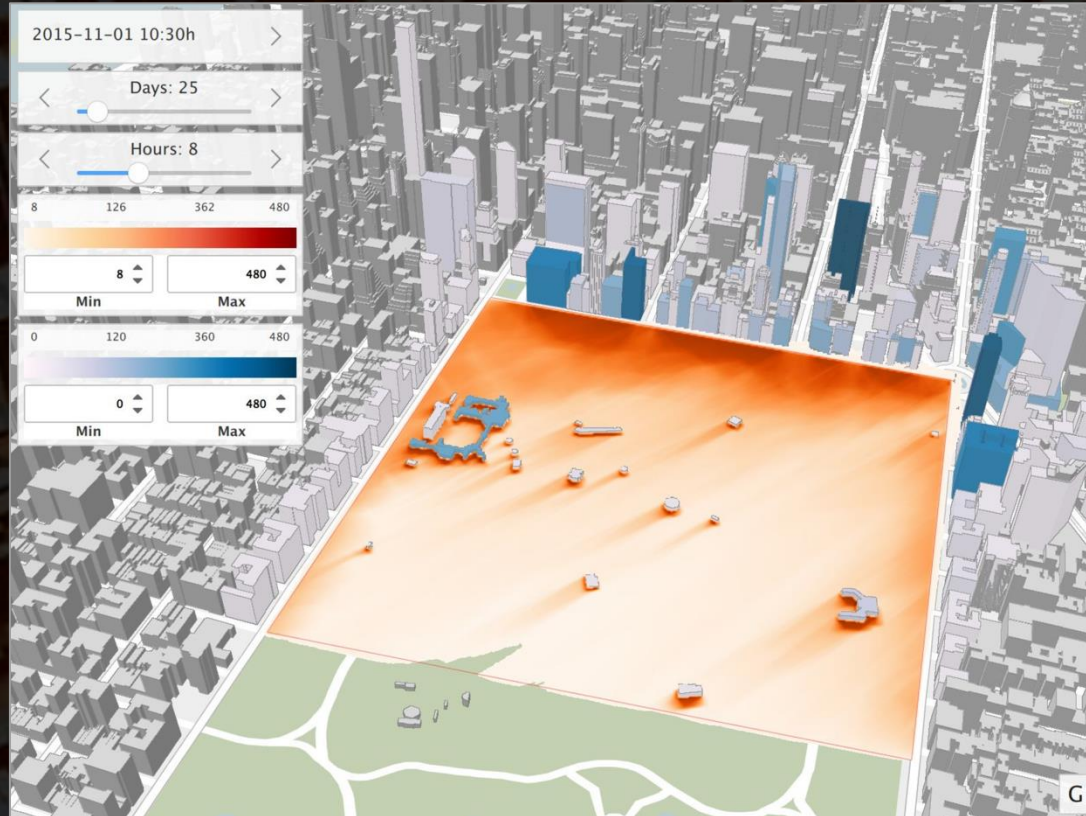
~2 hours time lapse

Mapping the Shadows of New York City - The New York Times

Quantifying shadow

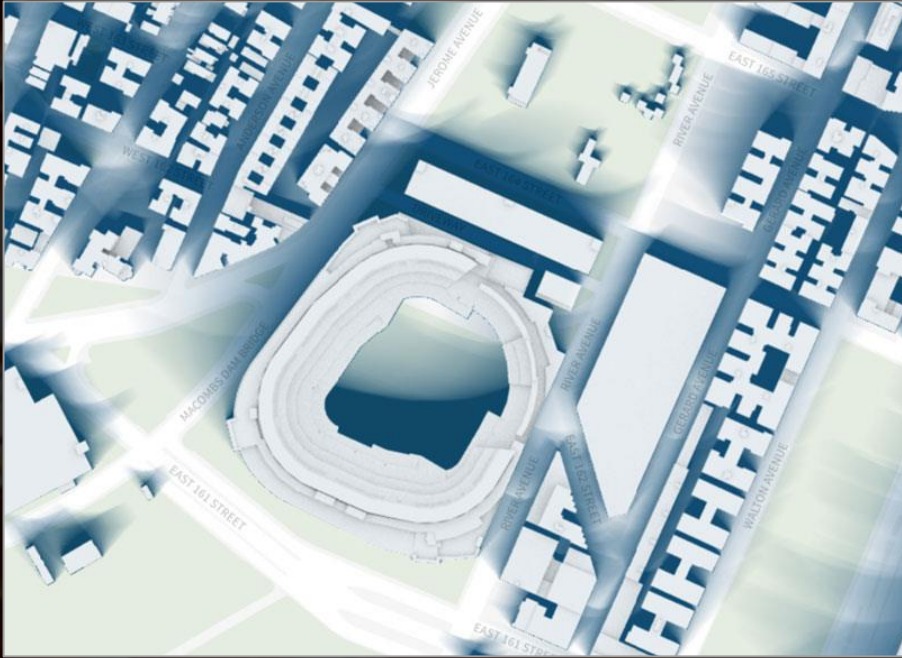


Shadow Profiler



Every Building, Every Block

Mapping the Shadows of NYC



Yankee Stadium



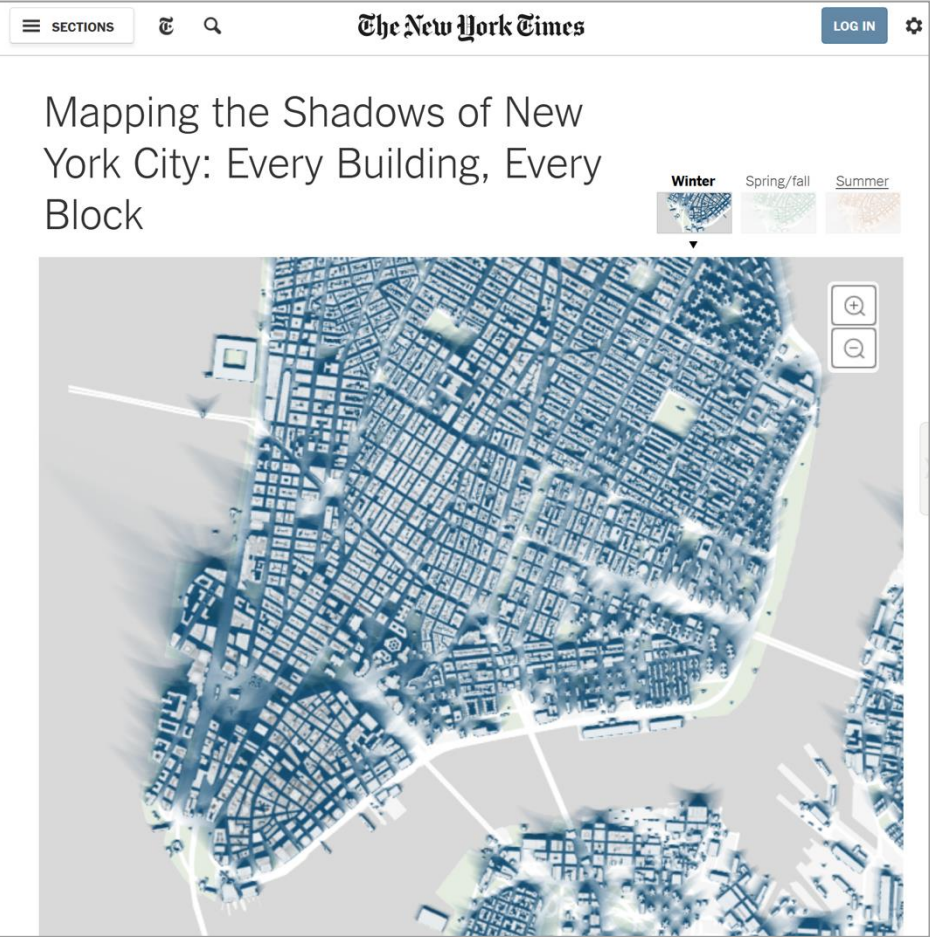
Financial District



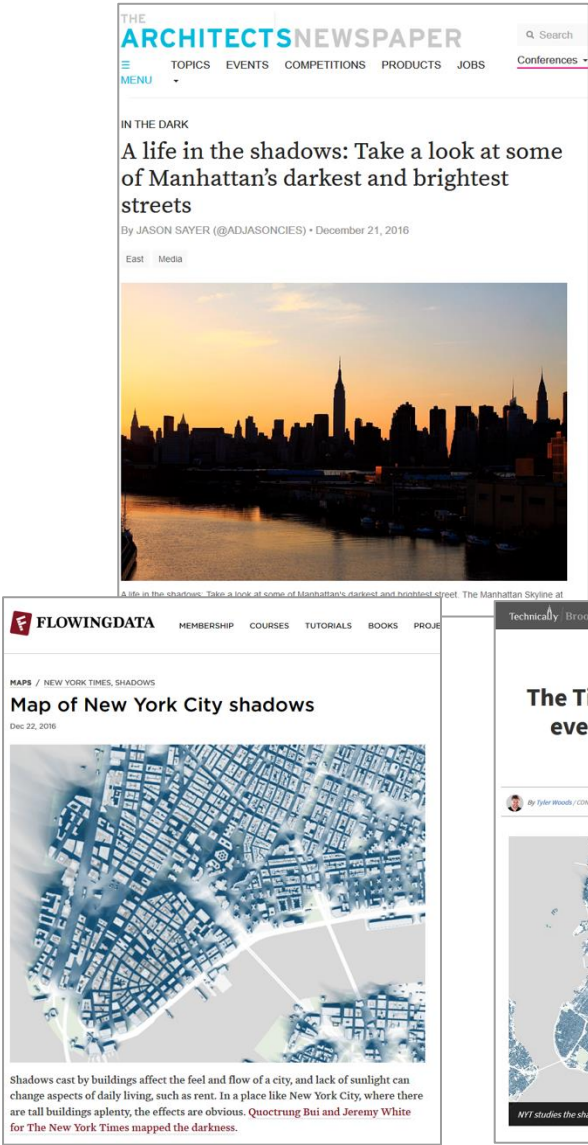
Broadway Avenue

Every Building, Every Block

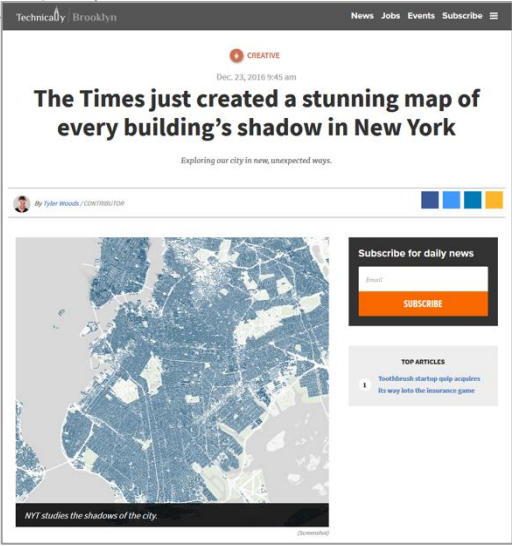
Mapping the shadows of New York City

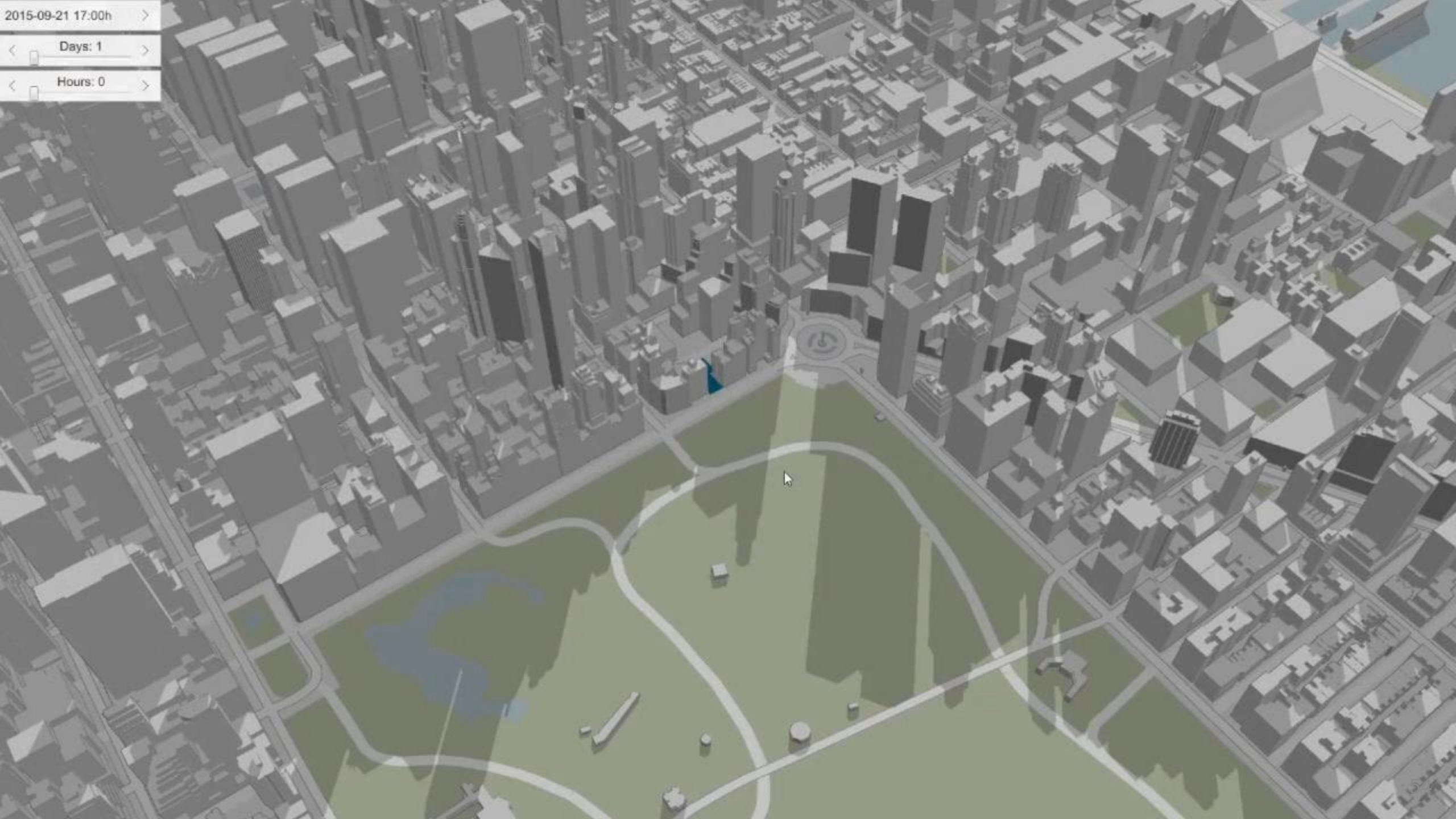


New York Times
<https://nyti.ms/2k0bF0G>



UC Berkeley





2015-09-21 17:00h

Days: 1

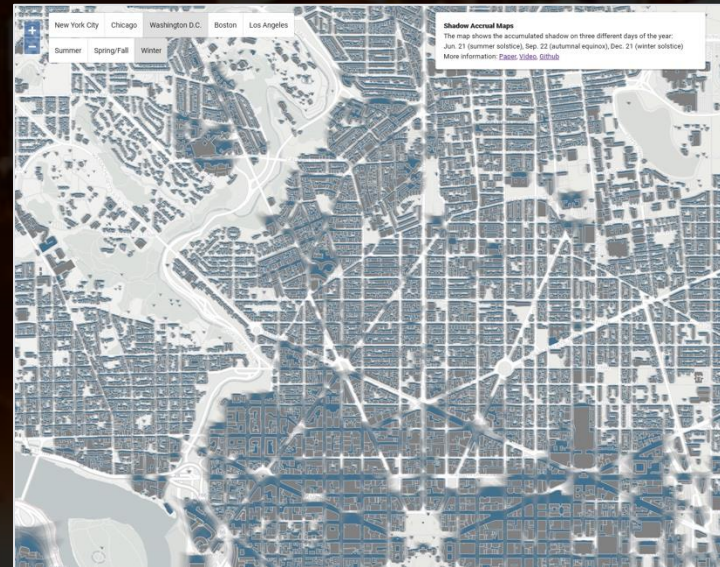
Hours: 0

Mapping the Shadows Across the US

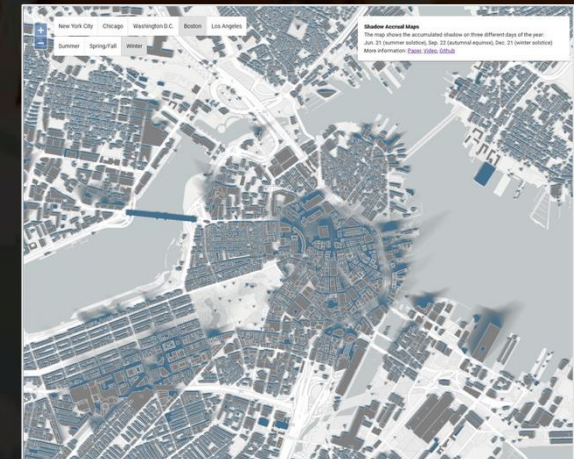
Chicago



Washington DC



Los Angeles



Boston

Every Building, Every Block

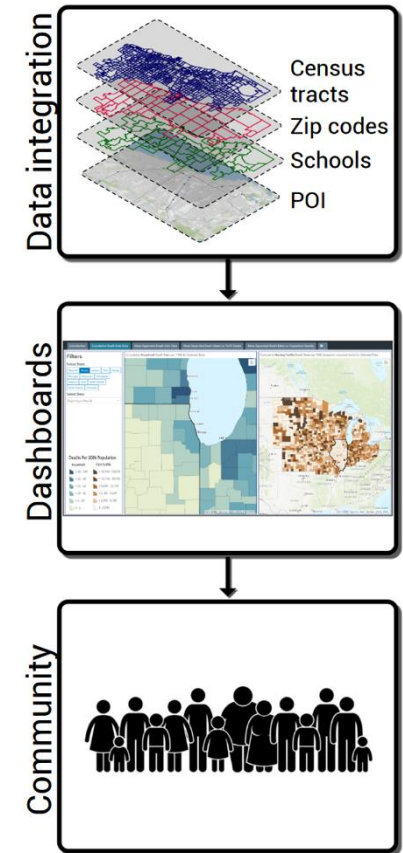
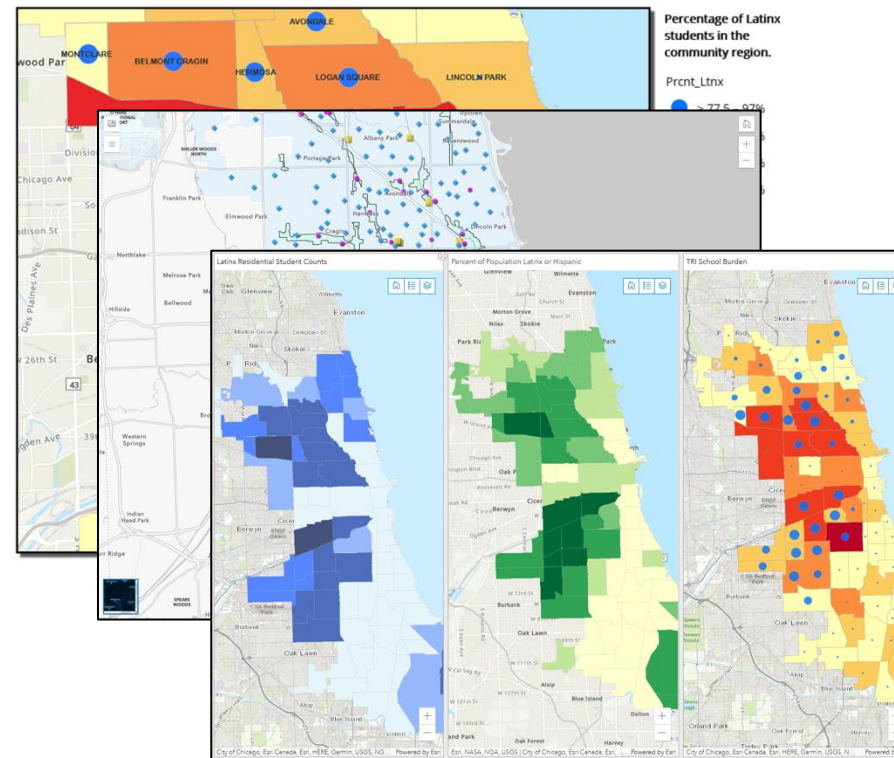
Air pollution

- Chicago ranks among the top five worst cities in the US for air pollution near public schools.
- Air pollution disproportionately impacts underrepresented communities in Chicago.
- Understanding and increasing community participation in environmental justice.



Midwest Comprehensive Visualization Dashboards

- Series of dashboards built in close collaboration with communities from Chicago.
- Six-month interaction period with community members.
- Assess cumulative impact of environmental hazards:
 - Industrial roads burden.
 - Rail yard burden.
 - Brownfield burden.



Urban Rhapsody: Large-scale exploration of urban soundscapes

Joao Rulff¹, Fabio Miranda², Maryam Hosseini¹, Marcos Lage³, Mark Cartwright⁴, Graham Dove¹, Juan Bello¹, Claudio T. Silva¹

¹New York University, ²University of Illinois at Chicago, ³Universidade Federal Fluminense, ⁴New Jersey Institute of Technology

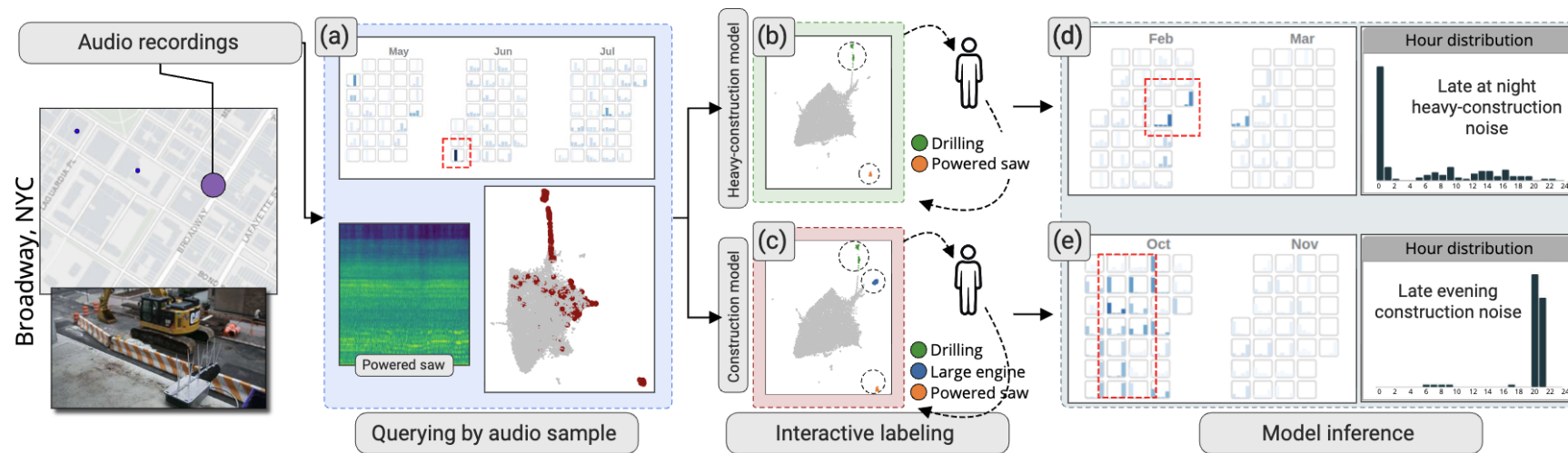


Figure 1: We use Urban Rhapsody to assess after-hour construction in New York City, first selecting audio recordings captured by sensors deployed around Broadway. Urban Rhapsody allows users to query using an audio sample, and drill down to days containing similar audios (a). Using the interactions provided by the tool, we are able to create classification models according to a user's perception of the soundscape (b,c), and then use these models to classify the entire data set and look for unusual events (d,e).

"Where Can I Park?" Understanding Human Perspectives and Scalably Detecting Disability Parking from Aerial Imagery

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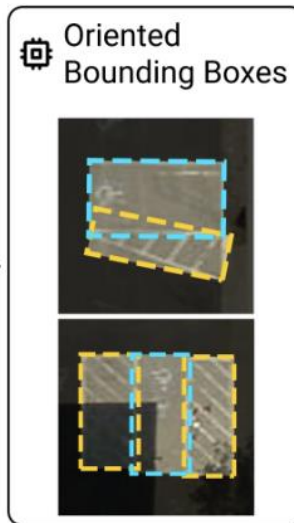
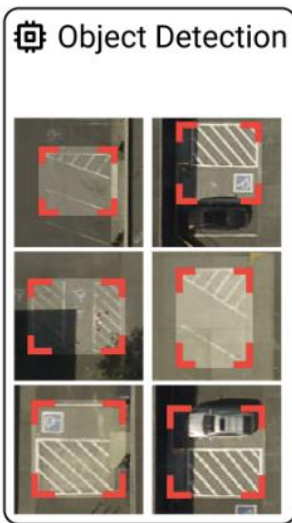
Allen School of Computer Science
University of Washington, USA
jonf@cs.washington.edu

Needfinding

PwD Interviews
(N=11)



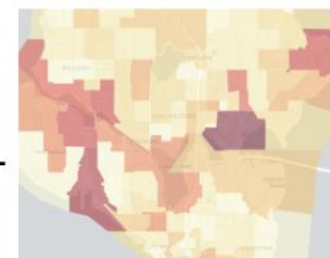
Orthorectified Tiles



Completed Detections



New Applications



Disability Parking Analytics



Disability Parking App

Needfinding

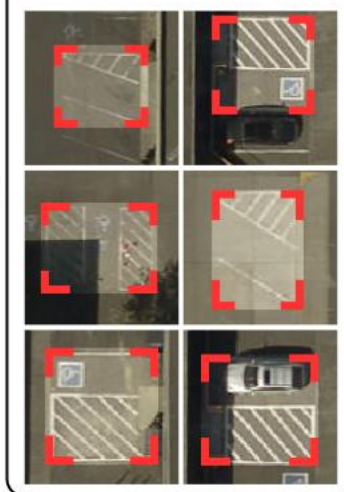
PwD Interviews
(N=11)



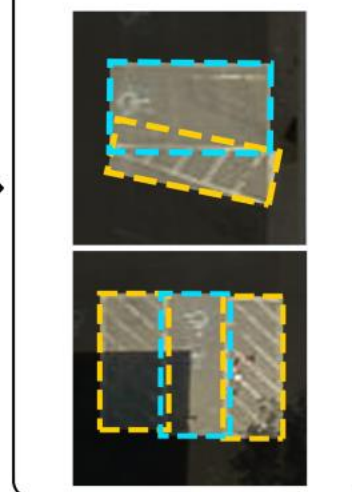
Orthorectified Tiles



Object Detection



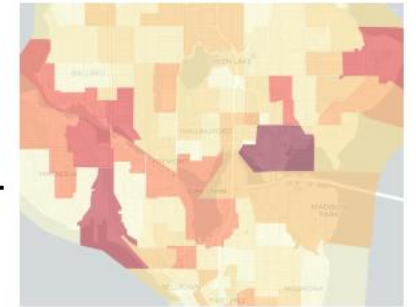
Oriented
Bounding Boxes



Completed Detections



New Applications



Disability Parking Analytics



Disability Parking App

Interview Study

Dataset & Pipeline Development

New Applications

→ Formative findings on how PwDs navigate disability parking, alongside policy recommendations

→ An open-source dataset of 11,762 labeled disability parking objects

→ A CV tool that locates, classifies, and characterizes disability parking from aerial imagery

LETTER • OPEN ACCESS

Urban and socioeconomic disparities in PM_{2.5} exposure across 340 Latin American cities

Edson J Ascencio, Antony Barja, Jose Montes-Alvis, Josiah L Kephart, Nelson Gouveia, Daniel A Rodriguez, Tarik Benmarhnia, Ana V Diez Roux, Usama Bilal, J Jaime Miranda

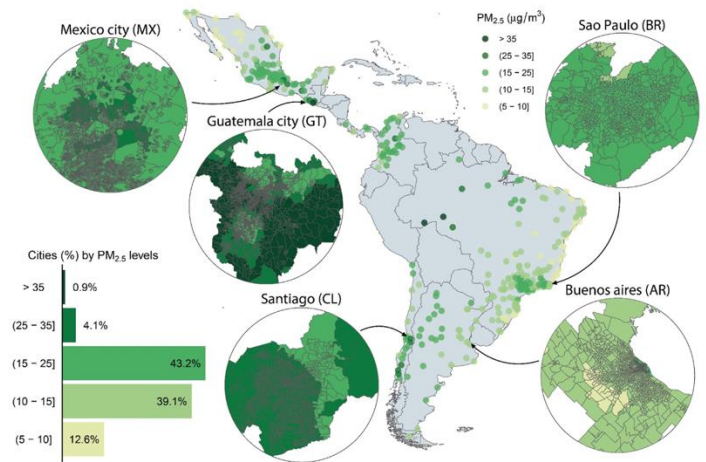
[Show full author list](#)

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[Environmental Research Letters, Volume 20, Number 12](#)

Citation Edson J Ascencio et al 2025 *Environ. Res. Lett.* **20** 124044

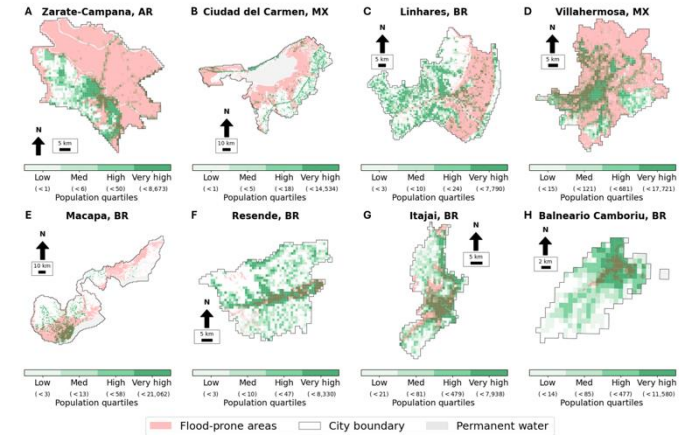
DOI 10.1088/1748-9326/ae20a4



Heterogeneity in Flood Hazard Exposure in 363 Latin American Cities

JINWEN XU , IRYNA DRONOVA, VICENTE TINOCO, JOSIAH L. KEPHART, USAMA BILAL, BRISA N. SANCHEZ, ANA V. DIEZ ROUX, NELSON GOUVEIA, OLGA L. SARMIENTO, [...] AND DANIEL A. RODRIGUEZ [+2 authors](#) [Authors info & Affiliations](#)

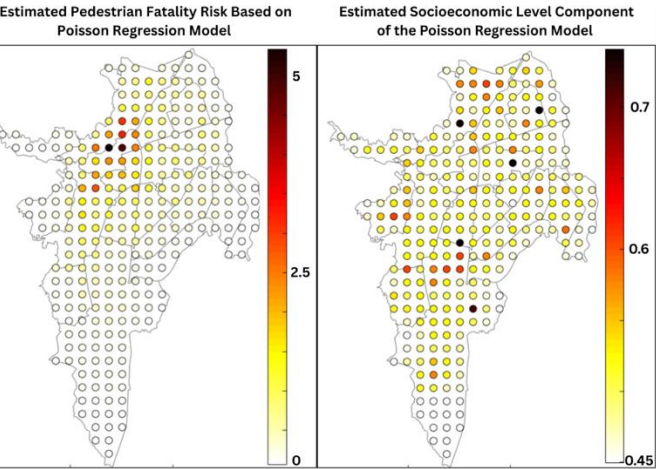
ECOSYSTEM HEALTH AND SUSTAINABILITY • 10 Jul 2026 • Vol 12 • Article ID: 0567 • DOI:10.34133/ehs.0567



Unequal risk: Longitudinal observation of socioeconomic disparities and pedestrian mortality patterns in Cali, Colombia

Andrés Villaveces , Lani Fox , Daniel A. Rodriguez, Shrikant I. Bangdiwala, Delia Ortega, Marc Serre

Published: August 10, 2026 • <https://doi.org/10.1371/journal.pgph.0006647>



Closing Last Mile Data Gap

**First Comprehensive Maps
for Food Aid Targeting in
Zambia &
The Republic of Congo**

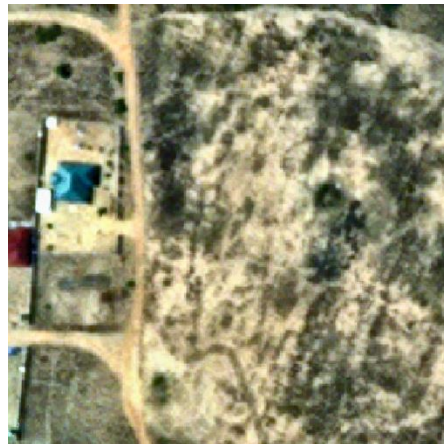


Haze Correction – No AI is needed!



CLAHE:
Contrast Limited Adaptive Histogram
Equalization

Localized contrast enhancement
boosts detail without over-amplifying
noise

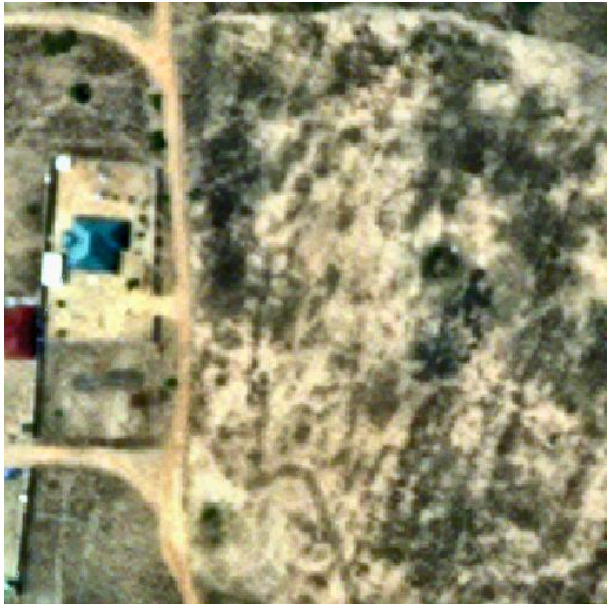


Closing the Last-Mile Data Gap

Original



Enhanced



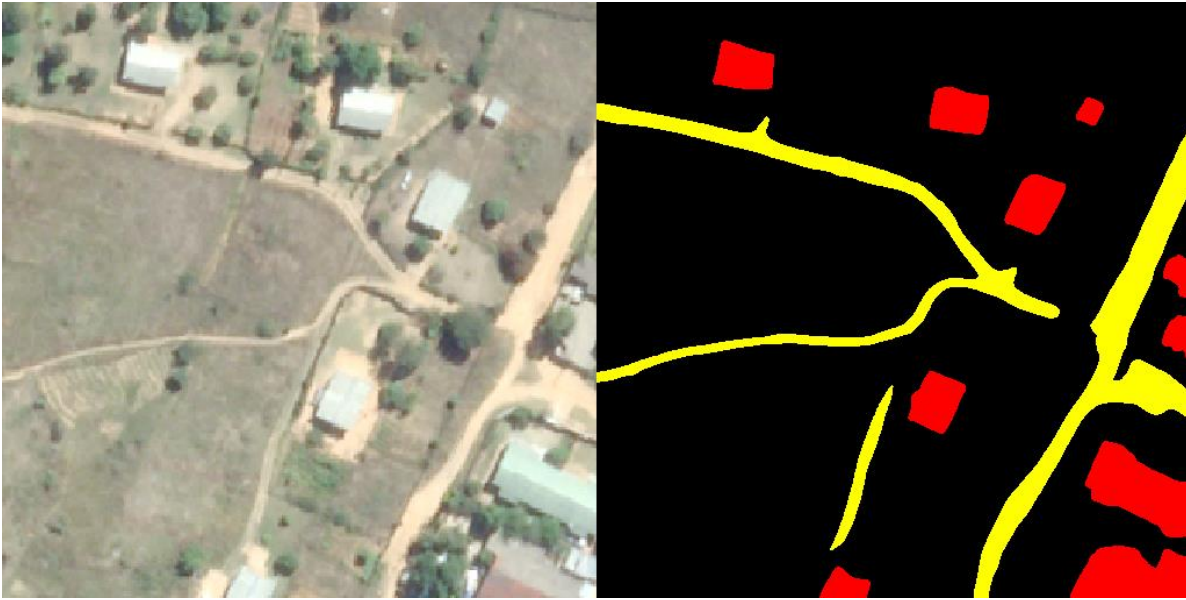
Ground Truth



Prediction



Work in progress – Preliminary results



Work in progress – Preliminary results

Democratize Access to Data

- Data collection impacted by:
 - Who is collecting the data?
 - Why this data is being collected?
 - How the data is being collected?
- We need open-source, Findable, Accessible, Interoperable, Reusable (FAIR) data
- As we are innovating, we are also setting norms, and we should be conscious and diligent about the norms we are setting

